

PRODUCT SPRINT / REVIEWER PACKAGE

InterviewMate - from CV intake to reviewable interview evidence

Case study, MVP requirements, phase reports and engineering handoff / 11 September 2026

This PDF is a readable submission export. The editable Markdown files remain the source of truth, and the production URL remains the runnable prototype.

Package	Current contents
Prototype	Production web app with recruiter pipeline, candidate interview and evidence review
Case study	Problem, UX, AI boundaries, scoring, evaluation limits and engineering handoff
Appendices	Phase 1-5 reports, MVP audit, release evidence, governance, roadmap, runbook and demo script

Submission status

The MVP is deployed and documented. The five-minute recording and final portal upload are still the remaining human submission steps. Manual-versus-assisted timing is disclosed as unmeasured rather than presented as a fabricated result.

Contents

Click a section below to jump directly to it. The first section is the concise case study and handoff; the appendices preserve the important current reports so a reviewer can trace decisions and evidence without opening the repository.

Quick access

[Open the production prototype](#)

[Open the interactive hands-on brief](#)

Main submission

1. Case study overview
2. Problem and user
3. Product decision and sprint contribution
4. UX and workflow
5. AI logic and scoring
6. Engineering handoff

Appendices

- [Appendix A - Product sprint brief and current status](#)
- [Appendix B - Phase 1 discovery and UX](#)
- [Appendix C - Phase 2 solution design and AI logic](#)
- [Appendix D - Phase 3 prototype build](#)
- [Appendix E - Phase 4 evaluation and iteration](#)
- [Appendix F - Phase 5 case study and handoff](#)
- [Appendix G - MVP readiness audit](#)
- [Appendix H - Production release validation](#)
- [Appendix I - CV pipeline validation](#)
- [Appendix J - Data governance](#)
- [Appendix K - Future feedback and privacy-aware learning](#)
- [Appendix L - Current runbook](#)
- [Appendix M - Five-minute demo script](#)

1. Problem and user

A tech recruiter needs to turn CVs and first-screen answers into a concise, job-specific record for a hiring manager. Separate resume checks, invitation setup, transcripts and assessment notes make the handoff fragmented. The recruiter must still inspect what a candidate actually demonstrated and identify unanswered areas.

The initial sprint focused on the transcript-to-brief bottleneck. The owner's later workflow requirements extended it upstream: upload a batch of CVs, inspect an ATS ranking, choose candidates and prepare interview links from the extracted identities. The evidence review remains the downstream purpose of that automation.

Primary user: recruiter managing the first-screen workflow for a technical vacancy. **Secondary user:** hiring manager inspecting the resulting evidence. **Affected user:** candidate who needs understandable questions, a chance to correct transcription, and a clear completion/evaluation experience.

The user problem is a desk-research hypothesis. No recruiter or hiring-manager interview was available. Discovery and the source register preserve the basis and limits. We do not claim observed applicant volumes, customer demand or time savings.

The intended success metric is active minutes to a reviewed first-screen brief, with unsupported claims, missing evidence and corrections recorded alongside time. Its before/after numerical baseline is not measured. The audit provides a small builder measurement protocol.

2. Product decision and sprint contribution

We reused an existing Next.js InterviewMate app with interview, authentication and resume-check foundations. Reuse allowed the sprint to focus on the recruiting workflow and AI behavior. This is an extension of an inherited product, not a claim that the full application was written in five days; actual working hours were not logged.

The delivered changes connect batch CV screening to invitation setup and interview history; combine voice with an editable draft and explicit Send; present evidence levels with exact candidate quotations; separate reviewer tests from production records; and add session recovery, feedback and deployed access controls. The inherited source baseline and phase reports provide the history.

MVP requirements and acceptance

ID	Requirement	Observable acceptance
R1	One role brief for a CV batch	Role/JD is entered once and used for each CV score
R2	Parse and rank CVs	Up to 50 PDF inputs produce parsed text or a visible failure; name/email are editable and successful rows show ATS scores
R3	Recruiter controls invitation selection	Top 5/10/20 is optional; manual checkbox changes determine who receives a generated link
R4	Separate schedule page	Steps 1-3 remain visible, including the empty ranking state; Step 4 opens /pipeline/schedule with selected candidates
R5	Snapshot invitation context	Link creation records identity, validity window, role/settings, CV context and available ATS result
R6	Candidate-only interview journey	Invited candidate can enter the permitted session, answer, view the allowed evaluation and leave feedback
R7	Voice and editable text together	Speech becomes a draft; only explicit Send submits the answer; draft clears after submission; interviewer speech completes before the next answer

ID	Requirement	Observable acceptance
R8	Grounded evidence assessment	Report shows candidate quotes and competency levels; invalid provider output is an error, not a fabricated successful assessment
R9	Reviewable history	ATS, assessment and timestamps remain inspectable on the session detail page in the same browser tab
R10	Separate test and production surfaces	/reviewer-tests and /invitations are secondary routes; primary history excludes reviewer tests
R11	Recovery and completion	Candidate sees interruption warnings/recovery or termination state; completion waits for closing behavior and offers the finish/countdown flow
R12	Feedback without affecting assessment	Five 1-5 experience metrics plus optional note; failed workspace sync is disclosed and a local copy can remain

These are the implemented product acceptance criteria, supported by the release checks and builder walkthrough. They are not independently executed pass counts for every browser/provider combination.

Out of scope: automatic rejection/hiring, automatic email delivery, ATS/HRIS integration, OCR, custom model training, covert app monitoring, eye tracking, a separate Figma app and a large multi-model tournament.

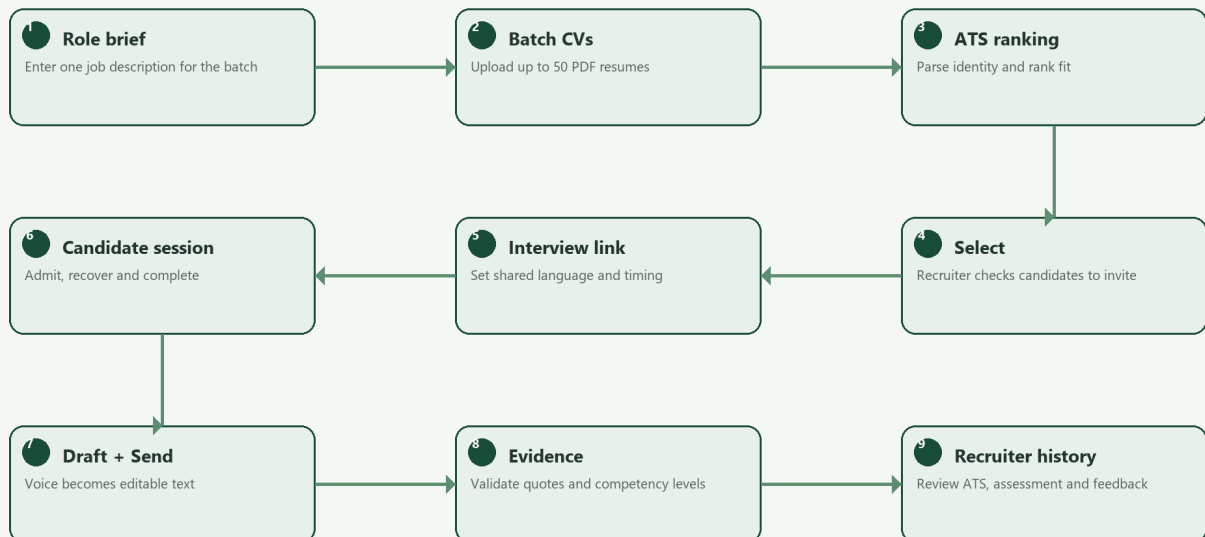
3. UX and workflow

The prior workflow was role definition -> separate CV review -> individual interview setup -> interview -> manual reconstruction of feedback. The MVP connects these stages while keeping recruiter selection explicit.

Visual flow

From CV intake to reviewable evidence

InterviewMate MVP handoff



Surface	Purpose and important states
/pipeline	Role, uploads, visible Step 3 empty state, per-file processing/error, ranking and selection
/pipeline/schedule	Selected identities, shared interview settings, validity and generated links; creation errors remain actionable
/apply/{sessionId}	Admission for the invited email and session window; denied/expired access must not open another session
/interview	Interviewer playback, bounded scrolling transcript, editable answer draft, Send, timer and recovery/completion state
/interviews and /interviews/{sessionId}	Production history with ATS/evidence/timestamps and full detail page; missing or pending assessments remain distinguishable
/candidates	Minimal candidate list with expandable screening/session information
/feedback	Admin inspection of experience ratings, separate from evidence scores
/reviewer	Private sponsored reviewer setup -> interview -> evaluation -> feedback
/reviewer-tests, /invitations	Secondary testing/admin routes, not primary recruiter navigation
/demo, /settings, /ats-check	Public demo entry, provider configuration where permitted, and standalone resume check

English is the target demonstration language; Bahasa Indonesia is the multilingual extension. Language selection controls speech/transcription instructions. Auto-detect is a separate configuration, not a claim of perfect recognition.

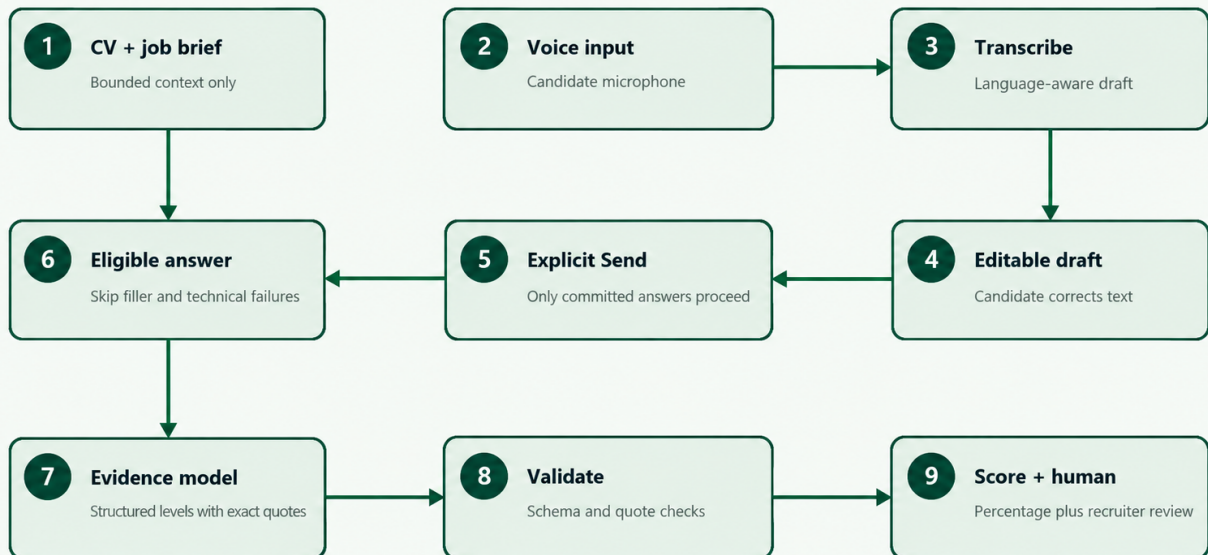
4. AI logic and scoring

Boundaries between deterministic logic and model output

Visual flow

AI boundaries and deterministic checks

InterviewMate MVP handoff



This is the behavior specification for the selected web-app deliverable. Transcription, conversation and evaluation have distinct responsibilities; this does not require a separate prompt-chaining product.

The release configuration records `gpt-realtime-2.1-mini` for voice, `gpt-transcribe` for transcription, and `gpt-5.6-luna` for evaluation. Low reasoning is the default, with supported Medium settings available. These are configured identifiers in this release, not a current market comparison. Gemini/DeepSeek adapters exist, but the recorded hosted smoke used OpenAI; an optional adapter is not a tested winner.

ATS score

The ATS route uses deterministic text matching, not an LLM:

```
overall_match = round(0.40 × keyword_match + 0.35 × skills_coverage + 0.25 × experience_alignment)
```

Each input component is on a 0-100 scale. Matching terms, missing terms, summary and heuristic flags accompany the score. The experience component uses coarse wording/seniority signals. Keyword wording, CV formatting and title language can affect it; the score is not verified job suitability, language-neutral matching or a validated probability of success. Recruiters can inspect it and override Top-N selection.

Interview and transcript behavior

The shared configuration holds language, duration, turn budget, approach, optional rubric, questions and technical panels. Role and validated CV context inform questions when custom questions are empty. Optional GitHub retrieval gives bounded context; a repository mention does not verify authorship. A short turn budget limits coverage, so a session is not guaranteed to ask about every CV item or repository.

Realtime instructions, turn policy and the interview store coordinate opening, explicit answer submission, repeats, follow-ups, recovery and closing. Prompt instructions request no duplicate greeting, unnecessary thinking narration or repeated answered detail. Those behaviors still depend partly on model compliance; observed failures belong in the review log.

Evidence evaluation

The assessment orchestrator passes bounded role/CV context, configured rubric and indexed transcript lines to the provider. The evidence prompt and validator request levels 0-4 with exact candidate quotations and 1-based turn references. Interviewer wording, filler, skipped and technical-failure turns are not candidate competency evidence. Transcript instructions are treated as untrusted content.

The server validates the structure and candidate quote references, then computes display dimensions. An exact quotation can still be assigned an inappropriate interpretation; mechanical validation does not establish semantic correctness.

- **Evidence level:** 0 = no evidence, 1 = weak/vague, 2 = partial, 3 = clear, 4 = strong/concrete.
- **Displayed evidence score:** $\text{round}(\text{sum}(\text{levels}) / (4 \times \text{total competencies}) \times 100)$ when competencies exist. Missing evidence contributes zero to this aggregate; it is not a separately imposed penalty.
- **Average evidence level:** average of assessed levels only, reported out of 4. Unassessed competencies do not lower this assessed-only average.
- **Coverage:** assessed competencies / total competencies. No derived/configured competencies can produce no score; an existing rubric with no eligible evidence can produce 0/100 with unassessed status.
- **Interpretation:** the display measures supported interview evidence under this rubric, not hiring probability or candidate capability. Different generated rubrics and different coverage make cross-candidate percentages imperfect comparisons.

The legacy `overallScore` hiring field stays null in the current evidence result. `dimensions.evidenceScore` is the displayed percentage. Engineers should not use the older `evaluation-schema.ts` hiring-recommendation schema as the current evidence contract.

The optional additional rubric defaults empty. Inspect the implementation when changing its semantics: the current evidence prompt restricts output to supplied competencies when a rubric is provided. It should not be documented as always adding those items to a separately generated base rubric.

5. Evaluation: inputs, results and iterations

Evidence source	What exists	What it supports
Authorized HR transcripts	Eight base cases and two variants, covering concrete/vague/missing/conflicting evidence, unknown speakers, injection, cautious wording and interruption	Reproducible inputs and provisional references; not calibrated hiring labels
CV/JD fixture	Ten fictional CVs, distinct names/emails and intended strong/partial/unrelated groups	Batch parsing, ranking inspection and invitation test material
Archived live benchmark	First frozen batch stopped at provider failure	Honest failure evidence; not a completed model comparison
Builder live testing	Owner acted as candidate/reviewer, supplied Indonesian transcript examples, reported bugs and accepted the revised flow	Single-builder qualitative product validation; not inter-rater reliability

Evidence source	What exists	What it supports
Release checks	191 tests / 25 suites, TypeScript/build and CI pass, production smoke	Software/integration evidence at the recorded release; mocked tests do not measure model accuracy
Seeded dashboard records	Fictional completed interviews and feedback used for demonstration	Presentation fixtures only; not real applicant outcomes or a satisfaction survey

The fixture data was authored for this project. No completed evaluation on an imported public HR dataset and no model training run is documented. English remains the intended baseline; no WER, latency distribution or independent English grading agreement is claimed.

Builder-observed failures and resulting changes

Observation reported during testing	Change made through the iteration	Evidence limit
Speech drafts reset/duplicated; noise or fillers advanced the interviewer	Editable draft plus explicit Send, transcript accumulation/deduplication and playback protection	Builder later accepted the interaction; no measured transcription error rate
Duplicate opening, spoken thinking filler, closing cut off	Opening/response coordination, instruction refinement, closing and finish/countdown control	User-reported improvement; probabilistic model behavior still needs observation
Transcript/footer overlap and oversized panels	Scrolling transcript and page spacing/layout adjustments	Qualitative desktop feedback; not a full accessibility audit
Evaluation pending despite a visible score; feedback permission/sync errors	Report/evaluation persistence, provider selection and feedback handling revisions	Builder acceptance and regression evidence; no fleet-wide reliability metric
Invitation creation failed on Vercel filesystem	Firestore-backed hosted ledger and stable secret configuration	Deployment/integration evidence in the release record

No measured recruiter time saving, model accuracy percentage, bias reduction or cost advantage is claimed. The narrow manual-versus-assisted baseline remains the explicit D1/D4 evidence gap. That requires measuring human work; more app features do not close it.

6. Engineering handoff

Source map and ownership

Concern	Current source
Pipeline and schedule screens	Pipeline/pipeline/page.tsx>), schedule/pipeline/schedule/page.tsx>)
Parse and ATS boundary	PDF result contract, ATS route
Session creation / snapshots	Firebase interviews
Voice and turn state	Realtime service, interview store, configuration
Evidence request / provider / validation	Evaluate API, assess, provider adapter, evidence contract

Concern	Current source
Report rendering	Evidence assessment
Admission and database policy	Access helper, Firestore rules, Storage rules
Hosted state	Reviewer access, ledger, Admin SDK
Experience feedback	Feedback form, admin view/feedback/page.tsx>)

API example and error contract

This is an **illustrative request**, not a recorded model result, for `/api/evaluate`:

json example

```
{
  "sessionId": "illustrative-session",
  "role": "Frontend engineer: React, API integration and testing",
  "cv": "Validated text from a fictional resume",
  "transcript": [
    { "role": "assistant", "text": "What did you personally implement?" },
    { "role": "user", "text": "I implemented the React filters and wrote tests for empty results." }
  ]
}
```

An optional configuration must match the shared schema. The result envelope contains `success`, `evaluation`, `provider`, `model`, `diagnostic`, `persisted` and `source`. This route currently returns `persisted: false`; successful generation is distinct from saving the session. Callers must check the appropriate persistence result before showing a saved state. Hosted demo/reviewer endpoints also perform their own admission and ledger operations.

This route bounds the request at 150 KB, transcript at 100 lines / 5,000 characters per line, role at 6,500 and CV at 24,000 characters. Invalid origin is 403, unknown provider is 400, and validation/provider failures are returned as 422 by this route. Error text does not create a valid score. Provider fallback uses configured permitted credentials; it must record the model that actually returned the output. Do not count fallback output as evidence about the failed primary model.

Persistence and access

`pipeline_candidates` holds screened identity/ATS metadata. `interview_templates` and `interview_sessions` hold role/configuration snapshots, identity, session status, ATS, transcripts and evaluation fields. The invited production CV can be stored under a scoped Firebase Storage path; public/personal demo context uses the relevant browser-local flow. Reviewer records are distinguished by `record_kind: reviewer_test` or `source: reviewer_invitation`; history filters must preserve that separation.

Feedback is attached as `candidate_feedback` on relevant session records, with a separate reviewer ledger path supported by the feedback form. Product feedback ratings must not change competency evidence. A local success with failed workspace sync must remain distinguishable from a confirmed server save.

Production hosted usage/invitations use Admin SDK Firestore transactions. The current ledger stores its state in one document: adequate for this prototype's bounded usage, but subject to document-size/contention limits. Local development/tests retain a file-backed fallback. Do not restore filesystem writes under Vercel `/var/task`.

The shipped UI exposes the recruiter workspace to authorized admin accounts; ordinary users see the public overview/demo/resume-check surfaces. Database rules also express creator and invited-candidate access. UI hiding alone is not the access boundary. Keep the owner account and temporary reviewer-admin access distinct; the recorded review account expires on 2026-09-18 at 00:00 UTC. Share credentials privately and renew deliberately if

the review happens later.

Run and maintain

Use Node 24, `npm ci`, then `npm run dev` and `http://localhost:3000`. Production uses Vercel and Firebase project `interviewmate-9bdd4`. Configure Firebase public application variables and server-only `OPENAI_API_KEY`, `FIREBASE_SERVICE_ACCOUNT_JSON`, `DEMO_COOKIE_SECRET` and `NEXT_PUBLIC_APP_URL` for the hosted path. Never place a service-account private key in a `NEXT_PUBLIC_` variable. Apply Firebase authorized-domain and rules changes to the matching project. See the current runbook for operations.

For code changes, use `npm test -- --runInBand`, `npx tsc --noEmit` and `npm run build`, plus targeted access/browser checks when changing those boundaries. For documentation, run `node docs/scripts/check-docs.cjs` and `git diff --check`. The recorded release checks were not rerun for this documentation-only audit.

Limits and next decisions

1. Measure a small manual/assisted review baseline and independently inspect candidate quotes/levels before making quality or productivity claims.
2. Validate scoring across role language, CV formats and missing-evidence conditions. ATS title heuristics and different generated rubrics can distort comparisons.
3. Add retention/deletion, access audit and stronger server-owned session state before broader real-candidate deployment. Browser focus events cannot prove cheating or inspect other devices; interruption accommodation needs human oversight.
4. Keep raw CV/transcript use purpose-limited. Cloud processing still transfers data to configured services; no candidate-data training pipeline is enabled.
5. Split the ledger or add infrastructure only when measured scale requires it. Additional providers, OCR and HRIS/email integrations are future work, not MVP submission tasks.

Future improvement: feedback-driven learning with local data preparation

The existing feedback form can support a reviewed learning loop: categorize recurring problems, select eligible examples, mask identifiers and normalize text in a private local worker, review the sanitized output, and create versioned regression datasets. Start with prompt/UX improvements; fine-tuning is an optional later experiment against a held-out baseline.

Candidate ratings indicate experience, not correct competency labels. Training targets require reviewed evidence. Local rules plus a small entity-recognition model are a practical first option; a locally hosted LLM can assist but does not guarantee anonymization. Names, contact information, rare career/project details and feedback comments all need inspection. Failed or uncertain sanitization blocks export.

This is a proposed pipeline, not a feature currently collecting training data. It requires a separate reuse decision, access/retention/deletion controls and human review. The current hosted interview already uses cloud processing; later local masking protects subsequent dataset preparation, not earlier transfers. See the future feedback and learning plan for the architecture, EN/ID validation, fine-tuning gates and withdrawal/model-retirement limits.

7. Submission package

Submit the production app as the **lightweight web-app** option, this Markdown as the **case study and engineering handoff**, and the finished **five-minute video**. The deployed app also includes a public interactive case-study reader at /case-study; it provides the quick narrative, phase tabs, workflow explorer, AI boundaries and links to the full PDF. A diagram supports explanation; no separate Figma or no-code workflow is required. The demo script covers intake, recruiter selection, candidate experience, evidence, testing and handoff in five minutes.

Video URL: not recorded yet. **Baseline values:** not measured. **Portal attachment/access check:** outstanding. The MVP works; these submission/evidence statuses must remain explicit.

Appendix A - Product sprint brief and current status

Supporting report included from the current repository documentation.

Current release status - 2026-09-11

Submission audit correction - 2026-09-11

The web-app MVP is demonstrable, but the literal D1/D4 manual-versus-assisted baseline comparison is still **not measured**. Synthetic fixtures, regression checks and builder self-tests are prototype evidence, not a completed independent model study. See the expected-output audit for each D1-D5 requirement.

The current case study and PRD now consolidates the engineering handoff, UX/AI diagrams, contracts, scoring and evidence limits. It supersedes the earlier PDF/JSON draft. The five-minute speech draft is ready; video recording, reviewer access verification and portal attachment remain outstanding.

Prototype format clarification: the challenge says **submit one** of its prototype options. We select the runnable lightweight web app. No separate Figma, no-code workflow or prompt-chaining app is required. Diagrams support the Markdown handoff.

Verdict: the prototype is ready for the portfolio demonstration and engineering handoff. Production is running at interviewmate-ai.shiverion.com with server-side OpenAI access, Firestore-backed reviewer/demo state, scoped recruiter records, and the completed voice + text interview workflow. The five-minute demo video is still a submission artifact to record; it is not represented as completed here.

Release evidence

Check	Result
Release branch / commit	main / fea1d2e
Automated tests	191 tests / 25 suites passed
Type checking and production build	Passed (npx tsc --noEmit, npm run build)
CI	GitHub Node 24 workflow passed: run 34555169845
Production smoke	Demo voice availability, AI configuration, hosted start, and Firestore usage ledger verified
Firebase	Rules, indexes, storage, and Admin SDK service-account access deployed for interviewmate-9bdd4
Deployment	Vercel production alias serves the public domain; encrypted OPENAI_API_KEY, FIREBASE_SERVICE_ACCOUNT_JSON, DEMO_COOKIE_SECRET, and NEXT_PUBLIC_APP_URL are configured

Phase readiness

Phase	Current result	Evidence and honest limit
1 - Discovery and workflow	Complete for desk research	Target user, bottleneck, workflow, concept, and baseline plan are documented. No recruiter time study was available, so the baseline remains a proposed measurement rather than a measured claim.
2 - Solution design and AI logic	Complete	UX flow, prompt/rubric behavior, provider configuration, recovery, guardrails, and data-governance decisions are implemented and documented.

Phase	Current result	Evidence and honest limit
3 - Working prototype	Complete	Production prototype supports batch CV/ATS ranking, invitations, voice + text interviews, editable transcripts, evidence evaluation, recovery, reviewer access, feedback, and scoped dashboards.
4 - Evaluation and iteration	Prototype validation available; baseline comparison incomplete	Synthetic inputs, regression results and owner acceptance exist. The original frozen live batch did not complete; manual/assisted timing and independent calibration remain unmeasured.
5 - Handoff and submission	Current Markdown handoff ready; video pending	Consolidated case study/PRD, diagrams and narration are available. Record the video and verify all three submission links/access.

Scope decisions for the submission

Future improvement: feedback-driven learning with local PII preparation. The existing form can inform issue triage; transcript reuse, local masking/normalization, curated datasets and optional fine-tuning remain proposed work with permission, privacy and held-out quality gates. No training on candidate history is enabled for the MVP.

- The primary baseline is English. Bahasa Indonesia is a separate multilingual extension result, not a replacement baseline.
- The automated CV pipeline is the product spine: one role brief -> batch CV parsing -> deterministic ATS ranking -> recruiter selection -> invitation links -> interview -> evidence-based evaluation.
- The product does not make automated hiring decisions. ATS and interview scores are review signals, with source-linked evidence and explicit human review language.
- Reviewer/demo access is hosted with server-side credentials and Firestore ledger state. Personal sessions can use BYOK; candidate data remains scoped to its owner/admin workspace.
- Local model hosting, model training on candidate data, eye tracking, OCR, exhaustive provider tournaments, and enterprise compliance are deliberately deferred to avoid overengineering the five-day prototype.

The dated production release record is the authoritative validation snapshot. Earlier reports remain below for traceability and are labelled historical where their pre-production results differ from the release.

<details> <summary>Historical pre-release audit and planning - superseded by the current MVP audit</summary>

Historical pre-release audit and lean finish plan - 2026-09-10

Historical snapshot: the following section was written before the final production validation. Its open items and older test counts should not be read as the current release status.

Verdict: not all requirements have passed. The software implementation is substantially built; live product evaluation and final submission are incomplete. This is an internal readiness assessment, not a verdict from the quest evaluator.

The original company challenge and our later 16 improvements are different scopes. The company asks for a useful working prototype, an honest case study/handoff and a five-minute video. It does not require enterprise infrastructure, custom-model training, a large model tournament or a statistically representative hiring study. We should now freeze feature development and finish the evidence and deliverables.

This section is the current priority order. The original challenge and all accepted improvements remain below for traceability. A deferred item is **not** being relabelled as passed. Earlier references to production readiness or independent reviewer calibration are engineering next steps, not additional company submission requirements.

Evidence checked

- Implementation commit: 5ac77a0 on codex/hr-product-sprint.
- Recorded validation: **138 automated tests / 18 suites; 24 selected browser checks; source lint; type checking; production build passed.** The dated record records the runs and their limits. This audit read those results and relevant source; it did not rerun unchanged tests.
- Model responses in automated tests were mocked. The no-answer assessment is deterministic and does not call a provider. These checks do not prove live speech quality, semantic grading accuracy or fairness.
- The seven-page PDF, editable content, video script and setup guide exist.
- Updated provider check: OpenAI HTTP 200 and one successful synthetic GPT-5.6 Luna evaluation; Gemini/DeepSeek host keys remain absent. See the model/access validation record. Human voice and comparative accuracy are not yet validated.
- No completed live model-comparison results, timed human baseline, final video or verified public reviewer deployment are recorded. The "0 of 3 attached" text later in this file was copied from the challenge page; it is not a live submission-portal status check.

Original five-day challenge: did we meet it?

Phase	What is ready	What prevents a complete pass	Readiness
1 - Disco very and workflow	Target recruiter, first-screen handoff problem, desk research, workflow, reuse decision and concept.	A measured baseline metric is missing. A small builder pilot can supply a clearly labelled baseline; access to a recruiter is helpful, not required to submit.	Mostly ready
2 - UX and AI design	Workflow/design artifacts, shared configuration, provider selection, prompts, evidence rubric, recovery and review flow.	Update the case study with any adjustments found during the live pilot.	Ready for prototype scope
3 - Working prototype	Next.js app, voice orchestration, bulk CV pipeline, ATS ranking, scheduling, evidence reports, reviewer access and tested UI/logic.	Demonstrate a successful real voice -> transcript -> assessment -> saved human review flow. Check the bulk scheduled path if included in the demo.	Implemented; live acceptance pending
4 - Evaluate and iterate	Synthetic cases, frozen historical data, regression tests, evaluation workspace, 10-CV pipeline fixture and result logging.	Actual model outputs, human judgments, observed failures and a manual-versus-assisted baseline comparison; bulk-pipeline live checks remain open.	Incomplete
5 - Handoff and submission	PDF draft, editable sources, organized documentation, pipeline handoff and five-minute script.	Insert measured pilot results, record the video and verify the reviewer can access the submitted prototype.	Incomplete

All 16 accepted improvements: current status

"Software-verified" means the relevant code/UI checks passed, not that every real-world behavior is proven. "Built; live check" means the feature exists but cannot yet be marked fully passed.

#	Requirement	Status	Smallest remaining check or honest limitation
1	Retry, return to setup and exit	Built; live check	Intentionally fail a real voice connection, recover, and confirm completed answers/configuration remain. Browser recovery behavior has passed.
2	Baseline and model comparison	Incomplete	Keep the frozen model/prompt/data baseline. Compare a small identical case set against one alternative first. A broad comparison across every provider is deferred, not passed.

#	Requirement	Status	Smallest remaining check or honest limitation
3	BYOK / Demo / Reviewer access	Built; host check	Verify one funded reviewer session and its limits. Demo limits identify a browser cookie, not a verified person; budget units bound operations, not exact currency spending. Retain basic caps and private keys; defer enterprise infrastructure.
4	Provider health and fallback visibility	Built; live check	Observe a real success/error and one controlled fallback. Do not claim fallback works across live providers based only on mocked calls.
5	Fillers, buffer, silence and mic failure	Built; live check	Speak fillers, pause/continue, wait for repeat/skip, and mute/disconnect the mic. Event/timing logic is tested; natural conversation is not yet measured.
6	Structured / Adaptive interviewing	Built; live check	Demonstrate a relevant follow-up without repeating answered detail and confirm the turn budget. Adaptive relevance currently relies on model instructions; a separate planning agent is unnecessary for this sprint.
7	Explicit evidence rubric	Logic verified; model grading pending	Empty input abstains deterministically, quotes are checked, and missing coverage does not lower assessed quality. Human-check real levels and whether the quotes actually support them.
8	One shared realtime service	Code verified; live check	Use the shared service successfully in the personal/reviewer room and a scheduled interview. There is no need for another transport rewrite.
9	Shared configuration and snapshots	Built; live check	Confirm a scheduled link retains its settings. Hosted Unlimited has a separate funding deadline; the public demo is deliberately limited. This is not literally unlimited hosted usage.
10	Prestige-neutral evaluation	Partially verified	Metadata invariance is tested, but identical answers with different employer/school/title labels inside the transcript still need a real model counterfactual check. Do not claim the model is bias-free.
11	Two-stage GitHub context	Built; sample check	Try one relevant public profile and one unavailable username. Check useful README context and non-blocking failure; no larger retrieval system is needed.
12	CV parsing and preview	Built; sample check	Try a text PDF, scanned PDF and invalid file; verify preview or explicit no-CV continuation. Error-string rejection is tested. Defer OCR.
13	Contextual review / synthetic separation	Software-verified	Navigation/access and synthetic reviewer storage are implemented. Verify the deployed reviewer path once; further product expansion is unnecessary.
14	Stable judgments and source-linked records	Software-verified	Browser tests cover typing, reload, submission and read-only completion; records contain source/model/rubric provenance. Collect a few actual human judgments next.
15	Prototype, PDF and video	Incomplete	PDF is a draft with honest gaps; video is a script only. Finish real test evidence, update the PDF and record the five-minute demo.
16	Documentation cleanup	Complete for this revision	Authoritative documents are identified, earlier designs archived, obsolete root reports removed and 504 checked links resolve. Keep final results synchronized.

Finish the quest without overengineering

1. **Make the existing primary stack work.** Configure a valid voice credential and one evaluator. Recheck access without exposing keys. Do not add providers or change models just to increase the model count.

2. **Run a small live acceptance session.** One English structured session, one English adaptive/recovery session and one short Bahasa Indonesia extension can cover the interaction checks above. Reuse the scheduled path for one English session. Record observed behavior and fix reproducible blockers only.
3. **Run a small synthetic evaluation pilot.** Reuse roughly five existing scenarios: concrete evidence, vague answers, missing evidence, contradictory ownership and an injection attempt. Also record the deterministic empty case, repeat a clear and a sparse case, and add a same-evidence prestige variant. Use the same rubric and keep original outputs, corrections and failures. This is a pilot, not a comprehensive benchmark.
4. **Human-check and time the work.** The builder may be the sole reviewer for this submission. For two comparable synthetic transcripts, time manually producing a brief and time verifying/correcting an AI draft; record the order, minutes, errors and learning effects. Label this a single-reviewer convenience pilot, not a recruiter productivity study. A second reviewer is a future improvement, not a blocker.
5. **Keep the optional model comparison small.** To address our added comparison requirement, run the same small case set through one alternative with prompt/rubric/data fixed. Record actual model, results, failure rate and elapsed time; use observed provider usage for cost if available. If access is unavailable, retain the primary and explicitly mark the comparison deferred. The original quest does not require all providers to be benchmarked.
6. **Finish the three artifacts.** Add actual pilot results and limitations to the PDF; record the five-minute video; provide a working reviewer-accessible prototype link or accepted runnable package. If offering hosted voice, verify its actual host and usage caps. Attach the artifacts and check access from a fresh browser.

The next action is **provider setup followed by the English live acceptance check**, not Phase 6 or another feature build. The runbook contains the detailed steps; use the smaller sequence above to prioritize them.

Current implementation extension - automated CV pipeline, 2026-09-11

The recruiter workflow now includes a focused automation from a role brief to scheduled interview invitations. This is an additive sprint extension built on the existing recruiter and ATS flows; it does not replace the evidence-based interview evaluation.

/pipeline accepts one role brief and a batch of up to 50 PDF CVs. For each file it parses bounded text, extracts an editable name and email, applies the existing deterministic ATS scorer, and sorts candidates by score. The recruiter may use Top 5/10/20 as a preselection shortcut, then must confirm the final checkbox selection before links are created. The score is a screening signal, not an automated rejection or hiring decision.

Creating a selected row's link snapshots the role/interview configuration and ATS result into the existing scheduled-session flow. The candidate signs in with the invited email, completes the same setup and live voice + text interview, and receives the allowed evaluation view. /candidates is the compact recruiter dashboard for all screened rows, including not_invited candidates, invited sessions and expandable parsed-CV/ATS/evaluation details. /interviews remains a compatibility route for older records.

The automation boundary is intentional:

- **Automatic:** PDF parsing, name/email extraction, deterministic ATS scoring, descending sort, status updates, invitation creation for checked rows, and configuration snapshots.
- **Recruiter-controlled:** correcting extracted identity fields, choosing or unchecking candidates, confirming the role and schedule, and sharing the generated links.
- **Not automated:** final hiring decisions, candidate rejection, sending email, model training, or continuous learning from candidate data.

Pipeline records are creator-owned. The pipeline_candidates Firestore rule, indexes and storage configuration are deployed in the production Firebase project; a denied read must surface as an availability warning rather than broadening access. Original PDFs are stored only for an invited scheduled session; uninvited rows retain bounded

parsed/ATS metadata. See the implementation note and the validation checklist.

Defer until after submission

- Additional LLM/transcription integrations or an exhaustive multi-provider, multi-language test matrix. The shipped defaults are `gpt-realtime-2.1-mini`, `gpt-transcribe` and `gpt-5.6-luna`; optional adapters remain available for a later controlled comparison.
- Local model hosting, custom model training, automatic candidate-data collection and continuous learning.
- Eye tracking, extra proctoring signals, OCR and more advanced GitHub retrieval.
- A separate adaptive-planning agent, broad analytics dashboards beyond the focused Candidates view, another frontend redesign and new report formats.
- Enterprise databases, distributed queues, extensive production compliance work and statistically powered fairness studies.

Keep the existing essentials: server-only host keys, reviewer admission, basic rate/usage caps, honest synthetic labels, safe recovery, human review and evidence-based abstention. If reviewers are given a publicly hosted voice demo, the current hosting/cleanup requirements still apply; "prototype" does not make exposed keys or unbounded spending acceptable.

Submission definition of done

- ☒ A fresh reviewer can access and finish the demonstrated workflow without supplying a personal API key; this was accepted on the production path.
- ☐ At least one recorded English live interview produces a usable transcript, evidence assessment and saved human review; the claimed adaptive/recovery behavior is shown working.
- ☐ A small synthetic pilot has actual outputs, human checks, failure notes and a clearly labelled baseline comparison. Indonesian results are reported separately if included.
- ☐ The PDF contains those actual results and honest limitations rather than only planned tests.
- ☐ The approximately five-minute video is recorded and playable.
- ☐ All three deliverables are attached/shared and their access is verified.

No overall completion percentage is assigned: more feature code does not compensate for missing evaluation evidence or a missing required video.

</details>

Original company challenge

Over 5 days of remote work, identify a real recruiting problem, design a near-future AI-native HR solution, build a functional prototype, evaluate its logic with mock data, and submit a portfolio-ready case study designed for engineering handoff.

Objective

Build a useful HR/Recruiting prototype that clearly answers:

- What recruiting bottleneck did you solve?
- Who had this problem?

- What is your near-future HR solution?
- How did you prototype the AI logic/UX?
- How do you know it works?

You may choose the problem yourself. The solution can be related to:

- Automated resume screening and scoring
- AI-assisted interview question generation for Hiring Managers
- Candidate matching algorithms and reasoning
- Conversational AI for candidate pre-screening
- Interview evaluation and feedback summarization
- Cold outreach automation for passive sourcing

The problem can be small. It must be real and highly relevant to modern recruiting.

Domains: Recruiting · HR · Talent Acquisition

Format

|||

|---|---|

| **Duration** | 5 days |

| **Schedule** | Monday to Friday |

| **Work style** | Remote (Expected commitment: 8 hours per day) |

| **Output** | Working prototype (no-code/UI+logic) + Case Study (Handoff doc) + Demo Video |

| **Company support** | Lightweight check-ins only |

Daily Rhythm

Day 1: Discovery & UX Workflow

Define the HR problem and the user's current workflow.

Expected output

- Target user (e.g., Tech Recruiter overwhelmed by 500+ inbound resumes)
- Problem statement and baseline metric (Time spent, error rate)
- Near-future solution concept
- 5-day scope

Day 2: Solution Design and AI Logic

Design the UX flow and map out how AI will solve the problem.

Expected output

- Wireframe or UX flow map
- Selection of AI tools (e.g., OpenAI API via Make/Zapier, V0, Lovable, Dify, or Figma AI)
- First draft of prompt logic or agent workflow

Day 3: Build the Prototype

Create a functional representation of the product. Remember, perfect coding is not required, but the logic and experience must be testable.

Expected output

- A working low-code/no-code workflow OR an interactive prototype with clear AI input/output specifications
- Core feature is demonstrable

Day 4: Evaluate with HR Data

Test your solution using synthetic data (e.g., 5 fake resumes, a mock interview transcript, or a dummy job description).

Expected output

- Test cases and results (Where did the AI hallucinate? Did the matching algorithm make sense?)
- UX adjustments based on test results
- Baseline comparison (e.g., "Manual screening takes 5 mins/resume; this prototype does it in 10 seconds with 85% accuracy")

Day 5: Engineering Handoff and Case Study

Prepare the final submission so an engineer and designer could take your work and build the production version.

Expected output

- Runnable prototype or interactive workflow demo
- Case study / PRD (Product Requirements Document)
- 5-minute Loom video

Deliverables

Working Prototype (The Near-Future Solution)

Submit one of the following:

- A no-code/low-code workflow
- A prompt-chaining app showing the AI logic
- An interactive Figma prototype coupled with a detailed AI behavior spec
- A lightweight web app built with tools like Cursor, Claude, or Lovable

Case Study & Handoff Document

Submit a Notion page or Markdown document covering:

- **The Problem & User:** Who are we solving this for in the HR space?
- **The Solution & UX:** How does the user interact with this feature?
- **Evaluation:** How did it perform with mock HR data? What were the limitations (e.g., AI bias)?
- **Next Steps for Engineers:** What does the dev team need to know to build this perfectly?

Demo Video

Submit a 5-minute Loom or screen-recorded demo showing:

- The recruiting problem
- The UX flow of your prototype
- The AI working in the background
- The test results and engineering handoff points

How Your Submission Is Evaluated

HR/Recruiting Domain Empathy

Did you target a real, painful bottleneck in the hiring process? Is the solution practical for Recruiters, HMs, or Candidates?

Product & UX Sense

Is the proposed workflow intuitive? Did you design an experience that makes the user's life easier, rather than just adding another clunky tool?

AI Prototyping & Logic

We don't need perfect code, but did you structure the AI logic well? Are your prompts, data flows, and agent behaviors well-thought-out?

Evaluation & Testing

Did you test your prototype with realistic HR data (resumes, job descriptions)? Did you identify where the AI might fail or show bias?

Handoff Readiness

Can our engineering and design teams look at your case study and immediately understand what needs to be built and why?

Submit Your Work

Every deliverable below is required - attach all of them to submit. Files can be a PDF, image, document or zip archive, up to 25 MB each. Your progress is saved as you go, so you can close this page and come back to finish.

Your Deliverables

0 of 3 attached

Working Prototype (The Near-Future Solution)

- [https://...](#)
- Add link
- No file chosen
- Upload file

Case Study & Handoff Document

- [https://...](#)
- Add link
- No file chosen
- Upload file

Demo Video

- <https://www.loom.com/share/...>
- Add link

Anything Else (Optional)

Extra files or links that do not belong to one of the deliverables above.

- Attach every deliverable above first. Extras unlock once your checklist is complete.
- No file chosen
- Finish your deliverables to add extra files

Add Links

Paste a GitHub repo, a live URL, a Loom demo, or any link to your work.

- Finish your deliverables first
- Label (optional)
- Add

Submit Quest

Accepted revision requirements - 2026-09-10

Things to Fix / Improve

1. Add interview recovery and safe navigation. Once a user enters `/interview`, there is currently no reliable recovery path if initialization, authentication, WebRTC, or an API request fails. Add `Retry Connection`, `Return to Setup`, and `Exit Interview` actions while preserving the user's interview configuration where possible so a technical failure does not force them to recreate the entire setup. A browser Back button should not be the main recovery mechanism; the application should explicitly handle failed initialization and allow the user to recover from the interview route safely.

2. Re-evaluate the current model stack instead of blindly upgrading it. The current OpenAI and Gemini models should be benchmarked against newer fast and cost-efficient alternatives rather than replaced simply because newer models exist. For realtime interviewing, prioritize latency, natural turn-taking, interruption handling, speech reliability,

availability, and cost rather than maximum reasoning capability. For post-interview evaluation, prioritize structured-output consistency, evidence grounding, repeatability, rubric adherence, latency, and cost. Establish the current system as a baseline first and then benchmark alternatives against the same evaluation dataset so model changes can be measured independently from scoring or product-logic improvements.

3. Implement three clearly separated access modes: BYOK, Demo, and Reviewer. BYOK users should be able to use their own provider credentials, and these credentials must never be persisted in Firestore, analytics, logs, or platform-owned backend storage beyond any temporary operation strictly required to establish a provider session. Demo Mode should use InterviewMate's server-side provider credentials with a normal usage limit such as five interview attempts per user per day. Reviewer Mode should use a server-validated invitation code that unlocks the application's full functionality, higher or apparently unlimited interview usage, Evaluation Sandbox, Human Review, and useful provider diagnostics while still enforcing backend expiration, revocation, rate limits, session limits, abuse protection, and a budget ceiling. InterviewMate-owned API keys must remain exclusively server-side.

4. Add provider health, active-provider, and fallback visibility. InterviewMate should understand whether each configured provider is Ready, Degraded, Rate Limited, Unavailable, or Misconfigured, and it should know which provider is primary, which one is currently active, whether fallback was triggered, and why. This diagnostic information should primarily be visible to recruiters and reviewers rather than candidates. Provider health should be derived from recent successful requests, actual provider errors, authentication failures, rate-limit responses, and cached server-side state rather than continuously sending unnecessary inference requests solely to maintain a status indicator.

5. Redesign realtime turn-taking, interruption, and silence handling. The current realtime conversation behavior is too sensitive, especially when the candidate says short fillers such as "uh", "hmm", or begins speaking without completing an answer. These events should not automatically cause the AI to interpret the answer as complete or advance to another question. The system should distinguish between filler-only speech, meaningful speech, interruption, thinking silence, completed answers, and microphone failure. After meaningful speech ends, provide approximately a 2-3 second end-of-turn buffer before the AI responds. For silence, allow a natural thinking period first, provide a gentle message such as "Take your time" after roughly seven seconds, and if silence continues for another approximately ten seconds, offer to repeat or skip the question before moving forward. A skipped question should be recorded as No Evidence Collected rather than directly penalizing competency, while microphone or technical failures must be handled separately and must never negatively affect candidate scoring.

6. Add optional adaptive follow-up interviewing. Recruiters should be able to choose between Structured Interview and Adaptive Interview before starting a session. Structured mode should follow the planned core questions, while Adaptive mode may ask evidence-driven follow-ups based on the current question, the candidate's answer, the competency being assessed, and remaining evidence gaps. Follow-up questions should explore relevant areas such as personal ownership, technical decisions, tradeoffs, validation, reasoning, outcomes, or missing detail, but they must remain directly connected to the current topic and must not ask for information the candidate has already clearly provided. Each follow-up should count as one interview turn toward the configured question budget, while competency coverage should be tracked independently from the number of questions or turns completed.

7. Redesign scoring around explicit evidence, competency coverage, and reliability. The current scoring system requires major validation because two equivalent no-answer tests produced materially different scores of approximately 40% and 60%, suggesting insufficient scoring stability and weak grounding in explicit metrics. Replace opaque percentage generation with an explicit competency rubric and separate dimensions such as Evidence Quality, Competency Coverage, Response Relevance, Consistency, and Assessment Confidence. Competency evidence could use a clear scale such as 0 = No Evidence, 1 = Weak or Vague, 2 = Partial Evidence, 3 = Clear Evidence, and 4 = Strong Concrete Evidence. Interview completion must remain separate from answer quality, so a candidate who reaches only five of seven questions but provides strong evidence can still score highly on assessed competencies, while unreached areas should be marked Not Assessed or Insufficient Evidence. If overall evidence is too limited to support a reliable conclusion, InterviewMate should return Insufficient Evidence for Reliable Overall Assessment instead of manufacturing a precise

score.

8. Fix and migrate the Scheduled Interview Realtime API flow. Scheduled interviews currently fail during realtime-session creation, and the larger architectural issue is that Demo Room, Scheduled Interview, and Reviewer sessions should not maintain separate realtime initialization implementations. Create one shared realtime service responsible for authentication, ephemeral credentials, session configuration, provider selection, WebRTC connection, fallback behavior, and error handling, and make every interview mode use that same implementation. This reduces the risk that one workflow silently remains on an outdated endpoint or configuration while another continues to work.

9. Unify Demo Room and Scheduled Interview configuration. Demo Room and Scheduled Interview should use one shared interview configuration schema and the same underlying defaults instead of exposing inconsistent settings. The shared configuration should include duration, maximum interview turns, Structured or Adaptive mode, competency rubric, preferred language, allowed interaction modes, custom questions, GitHub enrichment, technical or visual panels, and other interview behavior. Add duration options such as 5 minutes, 10 minutes, 15 minutes, and Unlimited, where Unlimited ends only when the candidate or interviewer manually ends the session or all configured interview objectives are complete. Follow-up questions defined in Adaptive mode should count as one interview turn, and scheduled interviews should snapshot their configuration at creation time so later recruiter-setting changes do not silently modify an interview that has already been sent to a candidate.

10. Make scoring evidence-based and prestige-neutral. Candidate scores must not directly increase or decrease because of company reputation, university prestige, job-title prestige, candidate name, gender, age-related wording, location, or other irrelevant attributes. Senior-level capability may still be evaluated when it is relevant to the role, but it should be inferred from demonstrated scope of responsibility, ownership, technical decisions, leadership, cross-functional influence, system complexity, and measurable outcomes rather than simply seeing a title such as "Senior Engineer" or a famous employer name. Add counterfactual fairness tests where two profiles contain identical interview evidence but different employer, university, or title labels, and verify that their competency scores remain equivalent.

11. Redesign GitHub enrichment into relevance-aware two-stage retrieval. The current GitHub enrichment already fetches real candidate profile and repository metadata, but repository selection is still too lightweight and the interviewer currently does not deeply inspect selected project content. Redesign this into a two-stage pipeline where InterviewMate first fetches lightweight repository metadata such as name, description, language, topics, stars, fork status, archived status, and recent activity, then ranks repositories against the job description and candidate context before selecting only the 2-3 most relevant or technically substantial projects. Only after that selection should the system retrieve README content when available, clean and bound that content, and inject only useful project context into the interview prompt. Job relevance, CV/project overlap, technical depth, relevant technologies, recent meaningful activity, and candidate ownership should matter more than raw star count, while forks, archived repositories, trivial tutorials, and irrelevant projects should be deprioritized. GitHub enrichment must remain optional, and enrichment failure must never prevent the candidate from starting the interview.

12. Add robust CV parsing validation and fail-safe behavior. Resume parsing currently extracts PDF text and uses it for ATS analysis and interview grounding, but parsing output is not sufficiently validated before being treated as candidate context. Parsing failures or server error strings must never be passed into the interviewer as though they were actual CV content. Replace the existing loose return contract with a structured parsing result such as success, partial, failed, no_extractable_text, or unsupported, together with extracted text, page count, character count, warnings, and failure reason. Before an interview begins, show the recruiter a short CV Context Preview or clear parsing status so they can confirm that the system actually read the resume. If the PDF cannot be parsed, the product should either ask for a replacement file or explicitly continue without CV grounding, and the AI must never fabricate resume-based questions. Image-only or scanned PDFs can initially be marked as having no extractable text rather than introducing rushed OCR during the five-day sprint.

13. Remove Review Brief from primary navigation and make it contextual. /review-brief should not appear as a normal top-level navigation item because it is an evaluation and human-annotation workflow rather than a primary recruiter or candidate destination. Normal application navigation should remain focused on routes such as Dashboard, Pipeline, and Settings, while Review Brief should be reached contextually from the relevant candidate, evaluation, or Reviewer Mode. Synthetic fixtures such as P1-A · Detailed account, P1-B · Conflicting ownership, and P2-SPARSE · Incomplete coverage should appear in a reviewer-only Evaluation Sandbox and be clearly labelled Synthetic Evaluation so they are not confused with real candidates. Synthetic evaluation cases must also be excluded from production statistics such as Candidates Screened, Average Score, Completion Rate, and Active Interviews.

14. Fix Review Brief state persistence and connect every review to its originating source. Selected review criteria currently risk becoming unchecked after the reviewer enters a reviewer ID or writes notes, which is a form-state bug that must be fixed because it undermines the validity of human evaluation. Review selections should remain stable while the reviewer types, adds notes, scrolls, or interacts with unrelated controls, and should only reset when a different transcript or fixture is loaded, the reviewer explicitly requests a reset, or a new review is intentionally started. Submitted reviews should become persisted or read-only completed records rather than visually reverting as though no review occurred. Every review should be linked to a specific candidate/session or synthetic fixture and record transcript or fixture version, rubric version, AI evaluation/model version, reviewer ID, criteria judgments, notes, and submission timestamp so human judgments can be used as reliable ground truth when evaluating the AI scoring system.

15. Design the submission and reviewer experience around three strong deliverables supported by clear internal sprint documentation. Keep the fully working Next.js application as the primary Working Prototype, including the realtime interview workflow, adaptive questioning, evidence-grounded evaluation, Reviewer Mode, Evaluation Sandbox, Human Review, and provider/fallback behavior, while keeping evaluation tooling hidden from ordinary navigation and exposing it contextually through Reviewer Mode. The primary Case Study & Handoff Document should be a polished self-contained PDF covering the recruiting problem and target users, the existing InterviewMate baseline, discovery findings, product and UX decisions, AI architecture, interview logic, evaluation methodology, synthetic test cases, results, failure cases, fairness and governance, technical limitations, engineering handoff, and next steps. The third deliverable should be the required approximately five-minute Demo Video showing the recruiting problem, interview setup, realtime and adaptive behavior, evidence-grounded evaluation, Evaluation Sandbox and Human Review, test results, limitations, and key engineering handoff points. Product_Sprint.md should remain the detailed sprint specification and source of truth containing the complete challenge context, requirements, findings, product decisions, architecture decisions, evaluation strategy, implementation scope, and deliverable plan, while docs/README.md should remain the living five-day progress tracker documenting completed work, current work, known issues, validation status, evaluation results, implementation changes, and remaining tasks. These Markdown documents should support rather than replace the reviewer-facing PDF, and separate PowerPoint or standalone HTML reports should not be prioritized unless the company specifically asks for them.

16. Clean up obsolete and conflicting Markdown documentation before submission. Audit all repository Markdown files and remove or archive documents that no longer represent the current sprint scope, architecture, product direction, evaluation methodology, or submission requirements. Files such as the original PRD.md, PROJECT_REPORT.md, outdated planning documents, duplicate reports, early architecture notes, temporary research files, or other Markdown documents that do not directly help explain the current MUST Product Sprint should not remain alongside the authoritative documentation if they may confuse reviewers about which version of InterviewMate is current. Product_Sprint.md should remain the authoritative detailed sprint specification, docs/README.md should remain the authoritative progress tracker, and the root README.md should remain a concise current product overview with clear links to the relevant sprint documentation. Historical documentation should only be moved into something such as docs/archive/ when it retains meaningful reference value; otherwise it should be removed entirely so the final submission repository presents one coherent and current version of the product.

Current implementation decisions - 2026-09-10

The accepted 16 requirements above supersede the original narrow review-brief scope. Reuse the Next.js baseline (commit 29e40ba) and its Firebase recruiter pipeline; do not rebuild from scratch. Preserve frozen historical evaluation data and prompts for independent model comparison.

- Primary user: recruiter preparing an evidence-backed first-screen handoff. Discovery is desk research; user pain and time savings remain hypotheses.
- English is the baseline; Bahasa Indonesia is a separately reported multilingual extension. A single builder reviewer is a pilot, not independent validation.
- Current defaults: GPT-Realtime 2.1 Mini voice, GPT-Transcribe (GPT-Live-Transcribe optional), GPT-5.6 Luna / Gemini 3.5 Flash-Lite / DeepSeek Flash evaluation. Optional Gemini 3.1 Flash Live uses OpenAI transcription and a server-held voice connection. Low reasoning is the default; Medium is available as documented. New choices are not an accuracy ranking.
- Recruiting data is creator-owned, with verified administrator access for miqbal.izzulhaq@gmail.com. Scheduled candidates sign in with the invited email and cannot write assessment scores. Firestore/Storage rules are deployed; old public CV tokens were revoked.
- Recruiter CV intake is available at /pipeline: upload a batch, parse names/emails, rank with the deterministic ATS signal, manually select candidates and create scheduled links. /candidates is the compact owner-scoped dashboard for screened, invited and not-invited rows; the latest pipeline_candidates rule must be included when rules are redeployed.
- Configuration v2 snapshots duration, turn budget, strategy, language, rubric, interaction modes, custom questions and optional technical/GitHub context. Hosted funding expiry remains separate from paused active time.
- Evidence rubric v2 exposes assessed quality, coverage, relevance, consistency and provisional confidence. No generated percentage or automated hiring recommendation. Exact candidate quotations are validated; relevance and semantic correctness still require human review.
- BYOK uses temporary request credentials; platform keys stay server-side. Reviewer invitations are signed, expiring and revocable, with rate/usage limits. Synthetic reviewer sessions use separate private storage.
- Personal data is not collected for training. Public GitHub context is optional and does not establish ownership. Production retention/deletion and infrastructure billing controls require engineering work.
- Deliverables: runnable Next.js prototype, self-contained PDF and approximately five-minute recorded demo. Markdown supports these artifacts. See current runbook, study protocol, live tracker, and demo script.

Current reviewer journey decision - 2026-09-10

The reviewer invitation is a constrained candidate journey. The administrator generates a signed invitation from Dashboard, shares it privately, and the recipient redeems it at /reviewer. The recipient can only set up the interview (role, language, shared voice/transcription settings, rubric, technical panel and optional CV), complete the live interview, review the evidence-based evaluation once, and finish. Voice + text is mandatory: the candidate speaks, edits the transcript draft when needed, and explicitly sends each answer. Reviewer recipients do not receive recruiter dashboard access, Evaluation Sandbox, Human Review controls or access to other candidates. The completed evaluation is saved in private server-side reviewer storage and surfaced only to the verified administrator's Dashboard.

This decision narrows the earlier Reviewer Mode wording in requirements 3 and 15 for the current prototype. Evaluation Sandbox and Human Review remain internal contextual tools for the administrator and sprint validation; they are not part of the link recipients use. The prototype still does not claim tamper-proof evidence, production

retention/deletion controls or hiring validity.

Appendix B - Phase 1 discovery and UX

Supporting report included from the current repository documentation.

> September 10 revision: see current progress and implementation matrix. Earlier entries below are a dated phase history, not a claim that the expanded scope has passed live acceptance.

Phase 1: discovery and UX workflow

Status: **Complete for desk research; manual timing baseline intentionally unmeasured.**

Submission clarification: the baseline metric is defined but its value has not been measured. This is a gap against the literal D1/D4 expected outputs, retained in the MVP audit. The current solution and broader CV-to-interview workflow are consolidated in the case study/PRD.

Updated: 2026-09-08 (Asia/Jakarta). Planned allocation: 8 hours; actual hours not recorded.

[Documentation home](#) · [Sprint index](#) · [Next phase](#)

Outcome

Reuse InterviewMate to explore **a recruiter-reviewed first-screen brief** for one fictional frontend-engineer role. The draft should connect claims to candidate turns, expose missing evidence and support correction before handoff. D01 owns the scope and unresolved design decisions.

The user chose desk research because recruiter/hiring-manager access is unavailable. This is enough to start a small design experiment, but demand, time savings, hiring accuracy and reduced bias remain unvalidated. Human timing is carried into Phase 4.

Release note - 2026-09-11

The discovery decision remains valid in the production prototype: the target user is a recruiter managing a first-screen queue, and the chosen workflow is role brief -> batch CV/ATS screening -> recruiter-controlled interview invitations -> evidence-based review. No recruiter access became available, so this report does not claim a measured manual baseline. The limitation is documented in the case study rather than treated as a product blocker.

Deliverables and reading order

Read	Artifact	Purpose
1	Problem and workflow	Target user, task, repository observations and hypotheses
2	Source register	Six external sources, alternatives and evidence limits
3	D01: sprint scope	Reuse decision, selected journey, scope and timeboxes
4	Discovery example	Short illustration of evidence versus interpretation
5	Evaluation materials	Practice records, source baseline and future human protocol

Verification

The source audit ran successfully; it inventories the inherited schema and report controls, not live AI behavior. Practice-material checks found 14 turns per case (643 words for P1-A, 673 for P1-B) and verified all 28 quoted reference snippets against their cited candidate turns. The references preserve self-report, testing limits and the ownership contradiction. These are internal artifact checks, not recruiter agreement; similar length does not establish equal difficulty.

Exit criteria

- [x] Establish desk research as the agreed discovery method.
- [x] Define a provisional user, problem, and job to be done.
- [x] Review primary sources with limitations and existing alternatives.
- [x] Map the existing app and hypothesized manual workflow.
- [x] Set a narrow concept and five-day scope.
- [x] Define baseline measures and a comparison protocol.
- [x] Explore one explicitly synthetic example.
- [x] Prepare two realistic practice records, separate source annotations, a blank brief, task instructions and an empty raw timing log.
- [x] Record a reproducible current-product capability inventory, clearly separate from human timing and AI-quality evaluation.
- [x] Resolve the discovery scope and hand the decision into Phase 2 through D01.
- [] Collect an empirical human timing baseline. No participant is assigned; this outstanding evidence is tracked in Phase 4, not marked complete.

Open items and next action

- Human timing is **not measured**; the study guide is ready when a human reviewer is available.
- Real workflow fit and provisional role criteria still need practitioner validation.
- The Phase 2 package now specifies screens, schemas, prompts, review states and acceptance rules. Product implementation is next in Phase 3.

Progress log

Date	Progress	Evidence / next action
2026-09-07	Reviewed the existing product and historical planning context	Reuse the interview/report foundation; keep this sprint distinct from legacy phases
2026-09-07	User selected desk research	No direct user findings are claimed
2026-09-07	Documented six external sources, alternatives, workflow and hypotheses	Category is established; target-user severity remains unknown

Date	Progress	Evidence / next action
2026-09-07	Prepared synthetic illustration, measurement protocol and scope	Next: realistic cases, human baseline when available, then detailed design
2026-09-07	Committed sprint documents on codex/hr-product-sprint as 0e9eba2	Six phase/index documents saved; inherited Firebase changes excluded
2026-09-07	Prepared full Phase 1 practice and measurement materials and checked source annotations	Two synthetic practice records; no participant session or prototype evaluation run
2026-09-07	Executed source inventory and finalized D01 discovery scope	Discovery package ready for Phase 2; human timing explicitly pending
2026-09-07	Organized research, decisions and evaluation materials into dedicated folders	Progress report now links to canonical artifacts; research conclusions unchanged
2026-09-08	Handed discovery into the completed Phase 2 specification	Practitioner validation and human timing remain pending

Appendix C - Phase 2 solution design and AI logic

Supporting report included from the current repository documentation.

> September 10 revision: see current progress and implementation matrix. Earlier entries below are a dated phase history, not a claim that the expanded scope has passed live acceptance.

Phase 2: solution design and AI logic

Status: **Complete - design package implemented in the production prototype.**

Current design handoff: the case study and PRD contains the shipped workflow/AI diagrams, percentage-score semantics and source-linked contracts. Earlier v1 design statements below, including no percentage display, are historical. The existing coded web app satisfies the selected prototype format; a separate Figma prototype is not required.

Started: 2026-09-07. Updated/completed: 2026-09-08 (Asia/Jakarta). Planned allocation: 8 hours; actual hours not recorded.

[Documentation home](#) · [Previous phase](#) · [Sprint index](#) · [Next phase](#)

Outcome

Defined InterviewMate Review Brief as four evidence-based criterion entries with traceable candidate quotations, explicit limitations, reviewer corrections and a reviewed export. D02 records the selected architecture: one structured server-side generation, deterministic checks and human review, with no numeric hiring score.

The clickable wireframe preserves the early design intent. The live product now implements the shared configuration, mandatory voice + text composer, evidence rubric, recovery/guardrail behavior, provider routing, reviewer boundaries and optional CV/GitHub context. Recruiter timing and independent model-quality calibration remain outside this single-builder release.

Release note - 2026-09-11

The design package is the source for the implemented UX and AI behavior. The current production validation is summarized in the release record; the older authored wireframe and design examples remain useful references, not a second product surface.

Deliverables and reading order

Read	Artifact	What is specified
1	Design package and wireframe	Complete artifact index, authored examples and how to run checks
2	UX and interaction specification	Input, source inspection, edits, review states, export, errors and responsive/accessibility behavior
3	Role and evidence rubric	Four provisional criteria, anchors, five shared questions and no score/ranking

Read	Artifact	What is specified
4	AI/data contract and exact prompt	Request/output shape, validation, model configuration, provenance, failures and review/export state
5	Acceptance plan	Software/UX expectations, eight-case evaluation plan, split and fixed readiness rules
6	Phase 3 backlog	Six ordered tasks within the eight-hour planning allocation

Verification and limits

Check	Actual outcome	Boundary
Offline contract checks	30 passed	Fixture/schema/source-copy checks and rejection cases; no provider calls
Wireframe interaction rehearsal	18 passed across wide/narrow DOM configurations	jsdom with dialog/media/scroll shims; not a real browser accessibility audit
Browser rendering	Wireframe opened in the Codex browser; narrow viewport screenshot inspected	Initial layout inspection, not recruiter usability validation or a full cross-browser walkthrough
Documentation links	291 local links across 36 Markdown files passed	All documents reachable; no external URL or factual-content validation implied
Source fixture preservation	P1-A/P1-B retain all 28 original turns and the five shared questions	Authored references remain provisional and known to developers
Semantic-overreach control	Authored unsupported claim passes mechanical checks as expected	Demonstrates why quote validity cannot replace human interpretation review

During wireframe review, source IDs were separated from hidden input-preview IDs and the source panel was kept mounted across narrow-layout edits. Interaction checks cover source highlighting/focus return, review gating, edit invalidation, removal/restore, empty evidence, provenance and failure/loading previews.

Reproduce the checks from the repository root:

powershell example

```
node docs/hr-product-sprint/scripts/check-phase-2.cjs
node docs/hr-product-sprint/scripts/check-phase-2-wireframe.cjs
node docs/scripts/check-docs.cjs
```

These results are design-artifact checks, not measured AI accuracy, time savings or candidate-assessment validity.

Exit criteria

- [x] Complete the flow and report wireframe.
- [x] Define four role criteria, evidence anchors and shared questions; label lack of practitioner calibration.
- [x] Finalize v1 input/output shape, prompt, provenance and failure/retry rules.
- [x] Select qualitative evidence status without numeric scores or hiring recommendations.

- [x] Freeze synthetic scenario expectations and readiness rules before live tuning.
- [x] Map the design to a bounded Phase 3 implementation backlog.

Open items and next action

The Phase 3 report records the implemented generation/review contract. The sentence about pending provider access belongs to the original Phase 2 boundary: Phase 2 itself used no API key or Firebase round trip. Subsequent production checks are recorded in the release record.

Human timing still awaits an available reviewer. The Phase 4 dataset now contains frozen C01-C08 inputs/references and two variants. They must remain versioned, with the withheld cases excluded from prompt tuning after authorship. Broader browser/accessibility checks and practitioner calibration remain future work.

Progress log

Date	Work completed	Evidence / consequence
2026-09-07	Created Phase 2 tracker and starting questions	Planning only; no completed wireframes or final AI contract
2026-09-07	Received Phase 1 decision, practice packet and capability baseline	Detailed design remained planned
2026-09-07	Defined D02, rubric/profile, prompt, executable contract, authored examples, UX states and acceptance plan	Versioned design choices and Phase 3 backlog; no live AI results
2026-09-08	Built the authored interactive wireframe and corrected source-panel/ID behavior	Source inspection, edits and review/export eligibility demonstrated with authored content
2026-09-08	Passed 30 contract checks and 18 DOM interaction checks; inspected browser rendering	Design package complete; Phase 3 implementation and Phase 4 evaluation remain outstanding
2026-09-08	Linked the implemented Phase 3 workflow	Design artifacts remain authored references; live provider verification is tracked in Phase 3

Appendix D - Phase 3 prototype build

Supporting report included from the current repository documentation.

> September 10 revision: see current progress and implementation matrix. Earlier entries below are a dated phase history, not a claim that the expanded scope has passed live acceptance.

Phase 3: prototype build

September 10 implementation: shared GA realtime service, three access modes with private reviewer invitations, one interview configuration, evidence-v2 reports and contextual persistent human review. Reviewer links now expose a constrained candidate journey (setup -> live interview -> one evaluation view -> done); evaluation tools remain internal. Voice + text is mandatory and the candidate explicitly sends edited transcript drafts. CV parsing and GitHub context have bounded validation/retrieval. See the current runbook and requirement matrix for implemented scope. Historical build details below describe the original review-brief baseline.

2026-09-11 release note: keyless dashboard access, provider settings, redesigned workspace, hosted voice, reviewer invitations, Firestore-backed usage state, and the batch CV pipeline are deployed at interviewmate-ai.shiverion.com. See the production release record. The pre-release access note remains archived for historical context.

Status: **Complete - runnable production prototype verified.**

Updated: 2026-09-11 (Asia/Jakarta). Planned allocation: 8 hours; actual hours not recorded.

[Documentation home](#) · [Previous phase](#) · [Sprint index](#) · [Next phase](#)

Outcome and deliverables

> **Historical verification below:** the original local Review Brief checks and failed-provider snapshots are retained for traceability. Use the release note above and the production release record for current readiness.

The local review workspace implements synthetic transcript input, four evidence criteria, exact candidate quotations, source inspection, corrections, review confirmation and JSON/text exports. It reuses the existing Next.js/React application, theme tokens and installed AI SDK. It has its own development-only page and API, outside the recruiter login/key gates.

The runbook and source map is the canonical setup and maintenance guide. The Phase 2 package remains the design reference; the backlog maps delivered behavior to B01-B06.

The current recruiter application also includes the bulk CV pipeline: `/pipeline` parses and ranks multiple resumes, the recruiter confirms invitation checkboxes, and `/candidates` keeps screened and invited records visible with expandable details. That path uses the existing scheduled-session persistence and is evaluated separately from this local Review Brief workspace.

The original **HTTP 503 / AI_UNAVAILABLE** response is preserved as a historical pre-release failure. The current release record documents successful production smoke and the passing automated checks. This still does not claim live model accuracy, recruiter productivity savings or fairness; those require a controlled study beyond the sprint prototype.

Implemented behavior

Area	Result	Main source
Input and role	Three practice transcripts, JSON import, fixed role confirmation, strict synthetic format and limits	Workspace, contract
AI boundary	Server-only key, pinned model, structured output, exact citation validation, one call per explicit attempt and 30-second deadline	Server adapter, handler
Evidence review	Full transcript context; both turns highlighted for conflicts; unknowns and follow-ups remain visible	Workspace above, styles
Corrections	Text/status/gap edits, claim removal/restore with stable IDs, preserved original, no editable source references	Review state
Attestation and handoff	Four checks plus reviewer ID; edits invalidate review; export only the current reviewed revision	Review state above
Failures	Sanitized errors and attempt IDs; invalid drafts rejected in full; explicit retry/cancel; stale responses ignored	Handler and workspace above
Provenance	Canonical input/role/prompt/contract hashes, source type, actual model metadata when available, original and reviewed export	Types, server adapter
Local boundary	No Firebase dependency for this flow; production page/API return 404	Page, API

Attempt records are local ignored synthetic generation records. Reviewer edits live in browser memory until explicit export; no Firestore writes or changes to the inherited scoring evaluator were introduced. The runbook explains exact records, hashes, input limits, filenames and how to keep runtime/design copies aligned.

Verification record

Executed on Windows with Node 24.11.1 and the repository's installed dependencies. Software checks use authored fixtures and injected/mocked provider behavior unless explicitly identified as the real attempt.

Check	Actual result	Practical limit
Default Jest discovery	36 tests passed across 3 suites: 32 new behavior tests and 4 inherited scoring tests	No model-quality claim
TypeScript	Application and separate Cypress projects passed	Separate Jest/Cypress matcher environments
Scoped ESLint	New feature, routes and browser spec passed	Full inherited application lint not claimed clean
Production build	Passed	Existing optional canvas/PDF rendering warnings remain
Development startup	Turbopack started; review page returned HTTP 200 with Firebase API key blank	Existing installed dependencies; no fresh install or full interview/Firebase flow
Production boundary	Actual GET page and POST API both returned HTTP 404 on the built server	Local production-mode check, not deployment verification
Browser workflow	7 browser tests passed in Electron: import, mock success/hash mismatch, source/edit/export, restore/invalidation, conflict/mobile, failure retention and cancel	Authored fixtures and mocked transport; not practitioner usability or model evidence
Real provider request	One attempt returned 503 / AI_UNAVAILABLE; no draft opened	Provider access must be repaired before a genuine live demo

Check	Actual result	Practical limit
Phase 2 preservation / documentation	30 offline contract and 18 wireframe checks passed ; all documentation links/anchors resolve	Historical baseline and practice source text remain separate

Meaningful checks cover byte limits, malformed/duplicate/no-candidate requests, invalid quotes/speakers, timeout/cancellation, explicit errors, provenance matching, source focus, edits, removals/restores, review invalidation and export fidelity. Unknown-speaker context is warned about and cannot support a candidate claim. A valid quote still does not prove semantic support; that remains a human review task. Source focus and return, narrow layout and export interactions were checked; a full keyboard/screen-reader audit, 200% browser zoom and other browsers were not tested.

Stabilization and implementation decisions

- Pinned Turbopack and output tracing to this repository, addressing the previously inferred parent-root startup problem.
- Excluded the nested local worktree from Jest/TypeScript/ESLint and split Cypress typing from Jest. No new dependency was required.
- Made the Firebase proxy tolerate React's `$typeof` inspection and changed the auth provider no-configuration message to a warning. Optional Firebase now produces explanatory warnings instead of false runtime errors.
- Kept the shared theme provider child tree stable while applying the saved theme; replacing its wrapper remounted the workspace and lost an early selection in the mobile browser check. The regression now passes.
- The new export names include transcript ID and revision. Content edits clear the relevant criterion check; reviewer identity and note changes clear all checks because the attestation has changed.
- The original local review-brief prototype kept voice, resume ranking, database persistence and the inherited seven-score reports outside that isolated path. The current recruiter application now has a separate bulk pipeline and candidate dashboard documented above.
- Retained the original failed provider record. Authored examples are an explicit separate action, never a fallback presented as AI success.

The repository's pre-existing Firebase configuration/rule edits and local agent work were left untouched. Inherited permission, candidate-entry and voice observations remain documented in the source baseline and local development guide.

Candidate session-integrity extension

Current behavior, 2026-09-09: pause/warning/recovery policy supersedes the earlier nonterminating defaults below. The full-screen modal pauses transport and time after one second hidden, warns on the second interruption or six seconds continuously away, and ends on the third or fifteen seconds away. Same-browser recovery retains answers/time and replaces the interrupted question. The current verification record separates mocked/software checks from unverified live voice and hosted persistence.

The following paragraphs record earlier iterations:

Added deterministic guardrails to the actual interview room: pre-start notice, active-only visibility/focus monitoring, a 10-second grace period, warnings at three counted events and human-review suggestion at five, without termination or score penalties. Candidates can provide structured context. Recruiter report display/export and a separate completion-save attempt are implemented; live Firebase persistence and anti-tamper guarantees are not established.

The local rehearsal requires no AI key. This is separate from the frozen synthetic review-brief experiment.

Verification record: 84 Jest checks passed across eight suites, including 19 guardrail checks; four guardrail browser checks passed. Build and scoped lint passed. The rehearsal returned 200 in development and 404 in production.

The alert follow-up adds a dismissible English/Indonesian pop-up for each new counted event and an optional local chime with preview/mute. The interview remains active. See alert behavior and manual checks.

The return-visibility fix removes the keyboard-focus prerequisite for closing a hidden interval. The rehearsal also shows monitoring off, a ten-second countdown, and readiness; Phase 4 records the reproduced regression and verification limits.

Exit criteria

- [x] Core feature runs with clearly labelled synthetic data.
- [x] Interactive prototype has a detailed AI behavior specification and a server implementation.
- [x] Reviewer can trace claims, identify unknowns and correct the brief.
- [x] Errors do not produce fabricated success states.
- [x] Setup and development/production boundaries are documented with executed checks.
- [x] Changed files and inherited functionality are distinguished.
- [x] Record the final browser/check results below.
- [] Demonstrate a successful genuine AI generation with retained provenance.

Open items and next action

Configure a server key with access to the pinned model, restart the loopback development server, and explicitly generate a new synthetic attempt. Keep its record alongside the failed attempt and inspect the claims for support. Phase 4 has now prepared its frozen dataset and runner, but its first live batch also stopped on AI_UNAVAILABLE. Do not report software pass rates as model accuracy.

Human timing and recruiter validation remain pending. A five-minute recorded demo and final case study remain Phase 5 work. No production deployment is included.

Progress log

Date	Work completed	Evidence / consequence
2026-09-07	Created build tracker and carried forward inherited risks	No new prototype behavior or repaired checks claimed
2026-09-08	Received completed Phase 2 design package and six-task backlog	Ready to build; authored wireframe and offline checks do not complete product implementation
2026-09-08	Implemented the separate local review workspace, API, validators, review state and exports	Runnable synthetic workflow; provider calls remain server-side
2026-09-08	Executed first real provider attempt	AI_UNAVAILABLE; retained original failure and did not substitute authored output
2026-09-08	Stabilized startup/discovery/typing and checked production boundary	Jest, TypeScript and build passed; production page/API returned 404

Date	Work completed	Evidence / consequence
2026-09-08	Finished seven browser checks and inspected desktop/mobile screenshots	Fixed theme-triggered early selection reset and versioned export names; authored/mockved evidence only

Appendix E - Phase 4 evaluation and iteration

Supporting report included from the current repository documentation.

> September 10 revision: see current progress and implementation matrix. Earlier entries below are a dated phase history, not a claim that the expanded scope has passed live acceptance.

Phase 4: evaluation and iteration

2026-09-11 release note: automated access/quota/provider-routing checks, the ten-CV pipeline fixture, owner acceptance and production smoke are complete. English remains the baseline and Indonesian the multilingual pilot. The production release record separates verified software behavior from model-quality claims.

Status: **Prototype validation documented; expected manual-versus-assisted baseline comparison unmeasured.**

2026-09-11 submission audit: D1-D5 audit and case study/PRD consolidate the authored transcripts, ten CVs/JD, builder self-tests, software checks and observed UX fixes. Prepared benchmark cases and seeded showcase records are not completed model runs. The MVP is demonstrable; the literal baseline-comparison expectation must remain open or explicitly disclosed as unmeasured.

Updated: 2026-09-11 (Asia/Jakarta). Planned allocation: 8 hours; actual hours not recorded.

[Documentation home](#) · [Previous phase](#) · [Sprint index](#) · [Next phase](#)

Outcome and deliverables

> **Historical runner details below:** the frozen dataset, old failure output and blank manual result fields are preserved as reproducible evidence. The current release status is the production validation record linked above.

September 10 revision: evidence, access, persistence and recovery checks are recorded in the revision validation. A separate model/rubric comparison protocol preserves baseline provenance for future work. The current release used owner acceptance and production smoke; it does not claim an independent accuracy, fairness or productivity benchmark.

The manual session-control validation checklist provides step-by-step rehearsal and live checks. The owner's self-test reports A1-A10 passing on the previous implementation, with warning-delivery/default-sound caveats and a visible-window focus gap. Part B is blocked by sign-in; a read-only Firebase check confirmed localhost is authorized but 127.0.0.1 is not. The resulting focus/sound changes require fresh manual validation and do not change the model-quality evaluation status below.

Phase 4 now has a frozen synthetic dataset, evaluation guide, repeatable runner and software checks. Eight base cases cover concrete evidence, vague accounts, missing topics, conflicting ownership, unknown speakers, instruction injection, cautious language and an interrupted interview. Two matched variants change an introductory name or conversational fillers.

The first genuine API batch attempted C01-r1 and returned **HTTP 503 / AI_UNAVAILABLE**; that result remains a historical pre-release record. Later production smoke verified the configured server path. The frozen runner and original failure are retained so a future controlled comparison can reproduce the limitation without rewriting the evidence.

Deliverable	Actual state	Evidence
Eight base cases and two variants	Authored and frozen before API calls	Dataset index, catalog
Expected behavior and split	32 base criterion judgments and 32 material unknown checkpoints; explicit sources and prohibited inferences	Reference files linked from the dataset index
Freeze/provenance	Canonical data hashes and exact runtime prompt/role/contract/model configuration retained	Freeze record
Repeatable evaluation runner	Plans 24 base + 6 variant runs; preserves attempts and stops on operational failures	CLI, shared harness
Real run evidence	One failed API attempt, zero valid drafts, 29 unattempted slots	Batch report
Semantic-review preparation	Per-valid-output templates implemented; none created because no output was valid	Evaluation guide
Software verification	27 checks passed using authored replays and mocked transport	Checker
Documentation navigation	401 local links across 40 Markdown files passed	All documents remain reachable from the root README
Human comparison	No human session; timing log still has no data rows	Manual guide, human log

The cases and expected judgments were authored by the project AI assistant, without recruiter/HM calibration. C07/C08 and their relevant variant are withheld from prompt tuning after authorship; they were seen by their author and are not independently blinded. No runtime prompt, role or product behavior changed during this phase.

Observed results and denominators

The machine summary and batch plan own exact run coverage. The attempt record includes both the sanitized HTTP response and the server failure record.

Measure	Actual result	Interpretation
Base attempts	1 of 24 planned	C01-r1 only
Variant attempts	0 of 6 planned	No sensitivity comparison available
Valid base drafts	0 of 1 attempted	Access failure; no candidate judgment produced
Available base criterion cells	0 of 96 planned	Status agreement is unavailable, not 0/96 accuracy
Proposed quotes / quote validity	No inspectable model output	No quote-validity ratio or hallucination conclusion
Full repeated base cases	0 of 8	Stability remains unevaluated
Matched variant pairs	0 of 6	No fairness or sensitivity result
Claim support, unknown preservation and conflict meaning	Not reviewed	Requires generated output and explicit source judgments
Human effort / time saved	Not measured	No human session

Measure	Actual result	Interpretation
Production readiness / hiring validity	Not established	A narrow synthetic study would not establish either

No failures were silently excluded. The one unavailable generation is counted as an attempted failure; the 29 untouched slots are listed separately. The earlier Phase 3 P1-A access failure belongs to its own smoke record and is not mixed into this dataset's denominator.

What the software checks establish

The 27 checks validate case/reference integrity, declared-only variant changes, freeze drift detection, exact-source constraints, input/configuration provenance, no hidden retries, and proper counting for partial or failed batches. Mocked complete batches verify that status disagreements and repeat/variant changes appear in the summary. Rejected quotes remain in the proposed-quote denominator when raw output is inspectable. Missing raw records stop collection rather than claiming a complete audit trail.

An intentionally unsupported statement with a valid quote passes the mechanical checker as expected. This demonstrates a validator boundary, not an observed model hallucination. The tests use explicit `mock_transport` metadata and write scratch data under ignored `tmp/`; they are not reported as 30 successful live generations.

Reproduce software checks without an API call:

powershell example

```
node docs/hr-product-sprint/scripts/check-phase-4.cjs
node docs/hr-product-sprint/scripts/evaluate-phase-4.cjs --check
node docs/scripts/check-docs.cjs
```

The guide owns live-run commands, stop rules, record maintenance and semantic-review instructions. The frozen acceptance plan still owns quality gates; software pass rates do not satisfy the model-quality gates.

Failures, changes and remaining uncertainty

The actual finding is operational: server key/model access is still unavailable. The evaluator's message intentionally does not expose provider error details, so the record does not distinguish rejected credentials from missing model access. One failed attempt was sufficient to stop this batch.

The harness now preserves that failure, reports the unattempted slots and leaves unavailable ratios unset. It also avoids sending expected labels to the model. No prompt tuning, AI-quality improvement, candidate ranking, or UX improvement based on live output is claimed. A content-failure-to-improvement sequence and any resulting UX change remain pending real generated outputs.

Do not modify the frozen v1 references after seeing outputs. Record disputes separately. If a withheld case informs a future fix, treat it as development and author a new withheld case. Practice P1-A/P1-B and Phase 2 examples remain development material.

Manual comparison

Condition	Completed human tasks	Active time median/range	Quality outcome
Manual brief	Not measured	Not measured	Not measured

Condition	Completed human tasks	Active time median/range	Quality outcome
AI-assisted, including source review and correction	Not measured	Not measured	Not measured

Use the manual study guide if a willing reviewer becomes available. Record real reviewer background, active work and AI wait separately. The source inventory, assistant execution speed and model latency cannot substitute for human task measurements.

Exit criteria

- ☒ Dataset, expected behavior and development/withheld split documented with actual versioned inputs.
- ☒ Human baseline unavailability disclosed clearly.
- ☒ Attempt recording, failure preservation and metric-denominator behavior verified.
- ☐ Complete planned base/variant run collection, or explicitly bound a reduced evaluation after working access is restored.
- ☐ Review evidence support, unknown handling, consistency and limitations on actual outputs.
- ☐ Record a meaningful content-failure-to-improvement sequence if content failures occur.
- ☐ Record and recheck any UX changes based on evaluation findings.
- ☐ Report draft quality separately from human-corrected quality once those records exist.

Open items and next action

Session-integrity software verification

Focus/sound follow-up, 2026-09-09: recorded checks passed 107 unit tests across 11 suites and eight browser scenarios, plus application/Cypress type checks, scoped lint and production build. Visible focus loss now participates in session control; overlap is deduplicated. Default-enabled audio repeats with faster triple pulses at final warning. This iteration responds to the owner's self-test; manual audibility/windowed behavior and live sign-in still need retesting. The prior automated iteration below remains historical evidence.

Current iteration, 2026-09-09: full-screen pauses and recovery now use one-second pause, second-interruption/six-second final warning, and third-interruption/fifteen-second ending. Recorded checks: 105 unit checks across 11 suites (40 integrity/control checks), seven browser scenarios, app/Cypress type checks, scoped lint and production build passed. Tests cover timing, terminal-state retention, offline/reload recovery, question retirement/replacement, optional sound, and mocked media stopping. Actual model speech/reconnection and hosted saves remain unverified. These are not model-quality or cheating-detection metrics. Earlier results below retain their original policy context.

The guardrail extension has recorded software checks: 19 new policy/store/hook checks and four browser checks passed; the full Jest suite now passes 84 checks. Coverage includes threshold boundaries, signal deduplication, continued interviews after five events, candidate context, storage failures, the actual interview start gate, and the Indonesian/mobile notice. Browser events and Firebase writes are mocked. Actual false-positive rates, recruiter interpretation, accommodation handling and live persistence need human checks; these results do not establish cheating-detection accuracy or complete the model-quality evaluation.

The session-alert follow-up adds five component checks covering silent defaults, sound opt-in/mute/failure, dismissal without clearing evidence, no replay from restored history, and Indonesian threshold messages. All five initial

guardrail browser scenarios passed, including keyboard dismissal and optional-sound controls. These exercise UI/audio calls; audible playback and interference during a real voice interview need human validation.

Return-visibility fix (2026-09-08): inspection of the user's Edge rehearsal confirmed the current alert UI was loaded and a pop-up was visible; this did not establish the cause of every reported missed switch. A new policy regression reproduced a delayed alert when the page became visible without keyboard focus (expected count one, received zero). It now passes after closing hidden intervals on visibility return. All 25 integrity unit checks and six guardrail browser scenarios passed, along with app/Cypress type checks, scoped lint and production build. The rehearsal now exposes startup countdown/readiness. The new browser regression simulates this event order; an actual embedded-browser tab-switch pass remains unverified.

English-first implementation verification

Recorded software checks: 65 distinct Jest checks passed (63 in the full suite, followed by a 13-check state rerun including two new ranking checks); 12 Cypress browser checks passed, including five benchmark tests. App/Cypress type checks, scoped ESLint, production build, and documentation links passed. Actual production GET page / GET API / POST API checks returned 404. The legacy 27-check harness and frozen dataset checks still pass. Optional inherited PDF/canvas build warnings remain.

These use authored mock provider responses. No live comparison, human reference approval, multilingual pass rate, hiring accuracy, or training improvement is claimed.

The English-first comparison workspace adds OpenAI/Gemini/DeepSeek adapters, reference approval, shuffled output review, separate language results, and export/import. English v1 is unchanged; three Indonesian adaptations need human meaning checks. This is a new experiment, not a replacement for the failed v1 batch. Configure a working provider privately, review C01, and perform one smoke run before expanding coverage. The governance record documents the synthetic-only input boundary and deferred training/transcription work.

For the legacy CLI experiment, restore server OPENAI_API_KEY access to its pinned model and restart the development server. For the new multi-provider workspace, follow the pilot guide above. Preserve the failed batch, conduct human semantic review, and report actual denominators before calling Phase 4 complete.

A final case study can already explain the scope, design, implementation and honest access limitation. Successful live AI, AI-quality measurements, human timing and the five-minute video remain unresolved deliverable work; no Phase 5 completion is implied.

Progress log

Date	Work completed	Evidence / consequence
2026-09-07	Created evaluation plan and empty result registers	No AI, fairness or time-saving results claimed
2026-09-07	Received practice materials and pending human-baseline task from Phase 1	Evaluation cases and actual runs remained planned
2026-09-08	Received v1 acceptance rules, rubric and run contract from Phase 2	Detailed plan now has one canonical source; dataset authorship and live runs remained outstanding
2026-09-08	Authored eight base cases, two exact variants and provisional references; froze them before API calls	69 base transcript turns, 32 base criterion judgments and 32 base uncertainty checkpoints
2026-09-08	Implemented and checked the evaluation runner and summaries	27 software checks passed; mocked outcomes remain explicitly separate
2026-09-08	Attempted the first real Phase 4 batch	C01-r1 returned 503 / AI_UNAVAILABLE; retained failure, zero generated drafts and 29 unattempted slots

Date	Work completed	Evidence / consequence
2026-09-08	Implemented English-first three-provider benchmark and a separate Indonesian extension	Reference approval, shuffled review, gated ranking, export/import, setup guide and governance record; software checks pass, live and human judgments pending
2026-09-09	Prepared the manual session-control validation checklist	Ten rehearsal checks, five live-flow checks, expected observations and a blank result sheet; manual results remain Not run
2026-09-09	Recorded the owner's manual self-test and iteration requests	Rehearsal results retain caveats; Part B blocked by unauthorized hostname; follow-up sound/focus behavior needs retest
2026-09-09	Clarified Part B entry/setup steps	Demo Room for initial B1-B3; scheduled sessions for B4/B5 saved-report checks; documented inherited UI/provider boundaries and remaining admission-clock gap

Appendix F - Phase 5 case study and handoff

Supporting report included from the current repository documentation.

Status: **Handoff document complete; video and final submission pending.** Updated: 2026-09-11 (Asia/Jakarta).

Planned allocation: 8 hours; actual hours not logged.

[Documentation home](#) · [Previous phase](#) · [Sprint index](#)

Outcome

Future roadmap addition: feedback and privacy-aware learning, linked from the PRD. The form is implemented; local sanitization, dataset curation and fine-tuning are explicitly future work, with no automatic use of existing transcripts.

Consolidated the shipped product into a case study and PRD, with workflow/AI diagrams, user requirements, acceptance criteria, scoring formulas, API example, persistence/access boundaries, source map and future work. The Markdown remains the editable source of truth, and the PDF submission package is the polished reviewer attachment generated from it.

The expected-output audit corrects the earlier broad "everything except video is passed" claim. The web-app MVP is demonstrable; literal D1/D4 manual baseline comparison remains unmeasured. The existing synthetic material and builder self-tests are valid prototype evidence but not independent calibration.

Engineering handoff checklist

- [x] Runtime/setup and secret variable names, linked to the existing release record/runbook.
- [x] Product and AI architecture diagrams.
- [x] Current API input example, output envelope and validation/error semantics.
- [x] Evidence rubric, percentage formula, quote/turn-reference rules and distinction from the legacy hiring schema.
- [x] Session snapshots, transcript/evaluation/feedback records and local/hosted boundaries.
- [x] UX routes, empty/loading/error and recovery/completion behavior.
- [x] MVP acceptance criteria and evidence limitations.
- [x] Bulk CV, ATS, manual selection and invitation handoff.
- [x] Access and privacy boundaries with no blanket security/compliance claim.
- [x] Operational limits and explicitly unmeasured latency/cost.
- [x] Future research and infrastructure work separated from MVP scope.

Final submission register

Required deliverable	Location	Status
Runnable prototype	Production app, runbook, recorded release checks	Available; confirm reviewer access window at submission

Required deliverable	Location	Status
Case study / PRD / handoff	Markdown source and PDF submission package	Created, exported and source-checked in this revision
Five-minute Loom/video	Seven-part English narration	Recording and final link not yet available

One prototype format is enough. The web app satisfies that choice; Figma, no-code tooling and a separate prompt-chaining app are not additional mandatory deliverables.

Recording plan

Time	Segment
0:00-0:30	Problem and target user
0:30-1:20	CV intake, ATS ranking and recruiter checkboxes
1:20-1:50	Interview setup and invitation link
1:50-2:55	Candidate voice, editable transcript and explicit Send
2:55-3:40	Evidence, ATS, history and feedback
3:40-4:25	AI stages, synthetic inputs, self-test results and limits
4:25-5:00	Handoff and next steps

Record separate parts if useful, then combine them into one video. Use a short actual English session excerpt and clearly label any prerecorded session or seeded result. Test accounts must use accessible emails rather than fictional example.com fixture addresses.

Exit criteria and next action

- ☒ Runnable prototype and existing release validation linked.
- ☒ Case study distinguishes inherited work, authored data, self-review, automated checks and unmeasured outcomes.
- ☒ Current source/contract and engineering handoff documented.
- ☐ Capture a measured builder baseline or retain its explicit absence; never substitute assumed time savings.
- ☐ Record, export and verify the approximately five-minute video.
- ☐ Check reviewer login expiry, link permissions and all three portal attachments.

Video URL: not recorded. Completing documentation does not mark the portal submission complete.

Progress log

Date	Work	Evidence / consequence
2026-09-07	Created handoff tracker	Outline only
2026-09-10-11	Prepared earlier PDF/script and documented deployment	Historical PDF retained; production release evidence linked
2026-09-11	Audited literal D1-D5 outputs; consolidated current PRD/case study and segmented narration	Markdown becomes canonical; baseline gap, single-builder evidence and missing video remain explicit

Appendix G - MVP readiness audit

Supporting report included from the current repository documentation.

Updated: 2026-09-11. Scope: documentation and source inspection, existing release evidence, and the builder's reported acceptance. No new live interview, recruiter study or model benchmark was run for this audit.

Challenge · Final case study and PRD · Demo script

Decision

The MVP is demonstrable and the web app satisfies the working-prototype format. Package the existing product rather than building additional prototype formats. The current case study/PRD is now consolidated in Markdown. Record the five-minute video and verify the submission links before handing it in.

There is one evidence gap in the literal D1/D4 expectations: manual-versus-assisted time/error comparison has not been measured. A baseline metric is defined, but its value is unknown. Earlier progress updates saying that every requirement had passed or that only the video remained were too broad. MVP acceptance, documented evidence, and completed submission are different statuses.

D1-D5 expected-output audit

Day	Expected output	Evidence available	Verdict
D1	Target user	Tech recruiter preparing first-screen feedback for an engineering hiring manager; discovery	Documented; proposed user segment, no practitioner interviews
D1	Problem and baseline metric	Evidence reconstruction and CV-to-interview handoff; active review minutes and correction count defined	Problem/metric defined; numerical baseline not measured
D1	Near-future concept	Batch CV intake plus recruiter-controlled invitations and cited interview evidence	Documented and implemented
D1	Five-day scope	Five separate phase reports; original app reuse disclosed	Documented; actual hours were not logged
D2	Wireframe or UX flow	Existing authored wireframe and current Mermaid flow in the PRD	Available; Figma is not required
D2	AI tools selected	Next.js, Firebase, OpenAI voice/transcription/evaluator and optional provider adapters	Source and release configuration documented
D2	Prompt logic or agent workflow	Turn controller, bounded context, structured evidence extraction and deterministic validation	Source-linked in the PRD
D3	Testable workflow with AI input/output	Deployed web app, input/output contract and error states	Available
D3	Demonstrable core feature	CV batch -> ATS ranking -> checked invitations -> interview -> report	Builder-accepted; production smoke recorded
D4	Mock HR cases and results	Authored transcripts, ten fictional CVs/JD, automated regression results and builder self-test observations	Available at prototype level; full frozen live benchmark not completed

Day	Expected output	Evidence available	Verdict
D4	UX adjustments from results	Transcript review before Send, duplicate-opening fixes, scroll layout, closing flow, feedback/report persistence fixes	Recorded builder feedback and reported acceptance; no independent usability score
D4	Baseline comparison	Manual workflow and test protocol exist; no measured before/after times	Evidence gap , not a missing product feature
D5	Runnable prototype or demo	Production app and source/runbook	Available; reviewer must use valid access
D5	Case study / PRD	Consolidated case study and PRD	Created in this documentation revision
D5	Five-minute Loom/video	Segmented narration and shot list	Script ready; recording and final playback check outstanding

Required deliverables: three, not six

The challenge says "**Submit one of the following**" for the prototype. These are alternatives:

Prototype alternative	What the format means	Our submission choice
No-code/low-code workflow	Executable Make, Zapier, Dify or similar workflow; diagram/PDF may explain it	Not selected; our app is coded
Prompt-chaining app	Testable app passing outputs through defined AI stages; accompanying prompt specification	Not a separate artifact we need to build
Interactive Figma + AI behavior spec	Clickable design prototype with explicit input/output and failure behavior	Not selected; no Figma artifact is claimed
Lightweight web app	Runnable application or hosted URL	Selected: InterviewMate

A PDF or diagram alone does not demonstrate executable AI behavior. Our Mermaid diagram documents the existing web app. The other two required submission items are **one case study/handoff document** (Markdown is explicitly accepted) and **one five-minute video**. A separate PDF export, separate PRD file and Figma prototype are optional. The combined case study/PRD covers both narrative and engineering requirements.

What data was actually used?

- **Authored synthetic transcripts:** eight base cases, two name/filler variants, 69 base turns and 32 base criterion judgments in phase4-eval-v1. Assistant-authored reference judgments are provisional. The archived first live batch failed after one request; its remaining runs cannot be described as passed.
- **Fictional CV/JD fixture:** ten distinct PDF CVs and a Senior Frontend Engineer role brief in cv-pipeline-v1, with intended strong, partial and unrelated groups. Expected bands are fixture design, not measured hiring accuracy.
- **Builder self-test:** the owner acted as candidate and reviewer, reported live interview/evaluation issues, and later accepted the corrected flow. This is useful qualitative product feedback, with one evaluator and no independent agreement metric.
- **Seeded interview/feedback records:** showcase data for screen demonstrations. They are not additional participants, successful live model runs or customer satisfaction measurements.

- **External data:** desk-research sources exist, but this audit found no documented imported public HR dataset used in a completed benchmark or training run. Do not say that a model was trained or validated on an internet dataset.

English is the intended baseline. The clearest conversation examples supplied by the owner are Indonesian. They support the multilingual extension; they do not establish measured English transcription accuracy. Capture an English interaction for the video and label it as a demonstration.

Smallest way to close the baseline gap

This needs human task performance, not another feature. Use two comparable synthetic transcripts and the same role/rubric. For the first, manually write the evidence brief before inspecting AI output. For the second, use the app and review/correct its brief. Time active work through a usable final brief; separately note generation wait. Record corrections, unsupported claims and missed evidence. Disclose that this is one builder, different cases and possible familiarity effects.

Measurement	Manual	AI-assisted
Case ID / role / reviewer / date	Not recorded	Not recorded
Active minutes to reviewed brief	Not measured	Not measured
Waiting time	Not measured	Not measured
Unsupported claims / missed evidence / corrections	Not counted	Not counted

Only calculate time reduction after measuring both values: $(\text{manual minutes} - \text{assisted minutes}) / \text{manual minutes} \times 100$. Do not call that percentage model accuracy or general recruiter productivity. If this small study is not performed, submit with the gap disclosed; do not mark the literal comparison requirement as fulfilled.

Submission checklist

- ☒ Working web app and current release evidence exist.
- ☒ Current Markdown case study/PRD includes UX, AI logic, evidence limits and engineering handoff.
- ☒ Five-minute speech draft and capture plan exist.
- ☐ Add a measured builder baseline comparison, or explicitly retain the unmeasured limitation in the submitted case study.
- ☐ Record the video, add its URL and check playback permissions.
- ☐ Open the prototype and case-study link as the intended reviewer; share any temporary login privately and confirm its expiry covers the review date.
- ☐ Attach all three required items in the submission portal. Its copied "0 of 3" text is not a checked portal status.

No additional MVP features are needed to perform these steps.

Appendix H - Production release validation

Supporting report included from the current repository documentation.

Progress tracker · Product specification · Current runbook

Status: **Prototype release accepted for portfolio demonstration.** The working prototype is deployed and the automated release checks pass. This record does not claim hiring validity, statistical model accuracy or independent recruiter validation.

Release identity

Item	Recorded value
Repository branch	main
Release commit	fea1d2e
Production URL	https://interviewmate-ai.shiverion.com
Vercel production deployment	virtual-ai-interviewer-assistant-aw2w1hcgx.vercel.app
Firebase project	interviewmate-9bdd4
Server-side production storage	Firestore Admin SDK ledger for hosted demo/reviewer usage and records
Production secrets	OPENAI_API_KEY, FIREBASE_SERVICE_ACCOUNT_JSON, DEMO_COOKIE_SECRET, and NEXT_PUBLIC_APP_URL configured in Vercel Production; values are intentionally absent from this repository

Automated validation

Check	Result
Jest	191 tests passed across 25 suites with <code>npm test -- --runInBand</code>
TypeScript	<code>npx tsc --noEmit</code> passed
Production build	<code>npm run build</code> passed
GitHub CI	Run 34555169845 passed on Node 24 with actions/checkout@v5 and actions/setup-node@v5
Firebase rules/indexes/storage deployment	Deployed to interviewmate-9bdd4; ownership and candidate-isolation rules remain enforced

Production smoke checks

- GET /api/demo/voice returned 200, available: true, five browser starts per UTC day and a shared host cap of 20 starts.

- GET /api/ai/config returned 200; the server reported the OpenAI evaluator as configured with gpt-5.6-luna. Gemini and DeepSeek remain optional and unconfigured in the hosted environment.
- /, /login, /demo and the API routes returned successfully over HTTPS.
- A hosted demo start completed successfully and persisted usage through the Firestore ledger. No local /var/task/.demo-state write was required.
- The reviewer invitation path was updated to use the same Firestore ledger and the configured server cookie secret; the former Vercel ENOENT: /var/task/.demo-state failure is fixed in commit e36010f and included in this release.

Product-owner acceptance notes

The owner walkthrough exercised the recruiter workspace, role setup, batch CV parsing and ATS ranking, candidate selection, invitation links, voice + text interview, editable transcript sending, evidence evaluation, reviewer-only access, feedback capture, recovery/guardrail behavior and scoped dashboards. Follow-up fixes from that walkthrough are represented in the current source and are covered by the automated suite above.

These observations are a single-builder acceptance pass. They are useful for a prototype demo, but they are not an independent usability study, model benchmark or proof that browser signals detect cheating.

Remaining submission work

1. Record and publish the approximately five-minute demo video using the demo script.
2. Regenerate the PDF case study if the recording adds new screenshots or measured observations.
3. Keep English as the baseline and report Bahasa Indonesia as a separate multilingual extension.
4. For a production hiring deployment, add server-authoritative interview admission/configuration, retention/deletion controls, accessibility accommodations, independent scoring calibration and a formal privacy/legal review.

Appendix I - CV pipeline validation

Supporting report included from the current repository documentation.

Status: **Implemented and owner-accepted in production; checklist retained as the repeatable regression protocol.**

The production release smoke verified the hosted pipeline dependencies, Firestore ledger and scoped access. The owner also accepted the recruiter flow from role brief through batch CV parsing, ATS ranking, invitation creation and candidate-only interview access. This file does not fabricate row-by-row benchmark measurements; a future recruiter study can fill the result sheet with a fresh commit/browser/account record.

This checklist validates the automation described in the bulk pipeline implementation note. It tests workflow correctness, schedule snapshots and access isolation. It does not measure ATS hiring accuracy, recruiter time savings or candidate quality.

Test setup

Use a fresh browser profile for the recruiter/admin and the fictional files in the cv-pipeline-v1 fixture. Deploy the current Firestore rules first, including the pipeline_candidates match. Keep the role brief in English for the baseline; repeat only the parsing smoke test with an Indonesian CV as a separate extension.

Suggested role: **Senior Frontend Engineer - InterviewMate AI**. The complete role text is in the fixture README.

Step-by-step validation

1. Sign in as the recruiter/admin and open /pipeline. Confirm the primary navigation says **Pipeline** and the old /interviews route remains available for compatibility.
2. Enter the suggested role title and job description. Upload all ten PDFs from dataset/cv-pipeline-v1 and choose **Analyze and rank resumes**.
3. Wait until every row has finished. Confirm each row has a non-empty parsed-CV preview, an editable candidate name, an editable email and an ATS score. Record any parse warning rather than correcting it silently.
4. Confirm the order is descending by score. The expected smoke-test bands are: Aisha Rahman/Bima Santoso/Clara Nguyen strongest; Dimas Pratama/Emily Carter/Farhan Yusuf/Grace Lim partial; Hendra Wijaya/Intan Maharani/Joko Setiawan weakest. Small ties or an unexpected order require inspection, not an automatic hiring conclusion.
5. Choose **Top 5**. Confirm only the five highest-ranked rows are preselected. Uncheck at least one preselected row and check one lower-ranked row. This proves Top-N is only a shortcut and the checkbox state is the final recruiter decision.
6. Correct one extracted name or email before creating links. Confirm the edited value is used in the invitation preview. Leave another candidate uninvited.
7. Configure a short scheduled interview, including language, duration, turn budget, Structured/Adaptive mode, rubric and any optional panels. Create links for the checked rows only.
8. Confirm every created session has the selected candidate email, role/configuration snapshot and ATS snapshot. Confirm the unselected row remains visible as not_invited and has no invitation link.
9. Open /candidates. Confirm the table is minimal by default and includes all ten screened rows, rank, candidate, email, ATS score and status. Expand at least one invited and one not-invited row; verify the detail includes

parsed-CV data, ATS evidence, invitation/session state, evaluation state and the saved configuration.

10. In a separate browser, open one invitation with the matching fictional email. Confirm the candidate sees setup -> live interview -> permitted evaluation -> done, and no recruiter navigation or other candidate data.
11. Try the same link with a different signed-in email. It must be denied. Try again after the session expiry; it must remain unavailable.
12. Return to the recruiter browser and confirm only the creator/admin can list the pipeline and candidate dashboard. A different recruiter account must not see these rows. Record a denied-read warning as a rules/deployment issue; never make the collection public to make the test pass.

Result sheet

Record the date, commit, browser, account role and exact sanitized errors before changing code.

Check	Expected	Result	Evidence / notes
1. Batch upload and parse	10 rows finish; non-empty previews or explicit warnings	Not run	
2. Name/email extraction	10 fictional names and emails populated and editable	Not run	
3. ATS ordering	Strong group above partial group above weak group	Not run	
4. Top-N preselection	Top 5 selects five rows only; manual edits persist	Not run	
5. Invitation creation	Links created only for checked rows; email appears without re-entry	Not run	
6. Snapshot integrity	Role, settings and ATS snapshot retained after setup changes	Not run	
7. Candidate dashboard	All screened rows visible; details expand without a wide default grid	Not run	
8. Candidate admission	Matching email can enter; different email and expired link cannot	Not run	
9. Recruiter isolation	Creator/admin sees rows; another recruiter does not	Not run	
10. Evaluation handoff	Completed invited session can be evaluated by recruiter/admin	Not run	

Interpretation rules

- A parse or score failure is a pipeline failure to investigate; it is not a reason to invent a candidate value.
- ATS scores are deterministic screening signals. They may help a recruiter prioritize review but must not automatically reject, rank for hiring, or select an interview slate without human confirmation.
- Link creation is not email delivery. The prototype creates and displays links; sending them remains a recruiter action.
- A successful software check does not prove voice quality, evaluation quality, fairness, legal compliance or production readiness.
- Keep results from the English baseline and Indonesian extension in separate rows. Do not pool them into one accuracy percentage.

Appendix J - Data governance

Supporting report included from the current repository documentation.

Pilot guide · Phase 5

Updated: 2026-09-11. Prototype policy and future requirements; no regulatory compliance claim. Current access and storage.

The candidate feedback form is implemented separately from the benchmark below. It collects five experience ratings and an optional comment; this does not automatically provide training authorization or ground-truth evaluation labels. The future feedback and learning plan proposes local PII masking/normalization, human review and optional fine-tuning. None of that learning pipeline is currently implemented or authorized to consume existing records.

Implemented

The benchmark accepts bundled fictional text only. It has no arbitrary transcript upload, audio collection, training job, automatic feedback collection, or automatic provider fallback. Do not substitute real candidate information into these fixtures.

The benchmark workspace holds its study in memory with explicit export/import. Review Brief and competency-v2 human-review forms also save drafts/completed records in browser localStorage; competency-v2 submissions with a valid invitation are saved in private reviewer storage. The model receives the selected transcript, rubric and instructions, not the human reference answers or judgments.

The recruiter CV pipeline is a separate, recruiter-triggered product flow. It stores bounded parsed text, extracted identity fields and the deterministic ATS snapshot in the creator-owned pipeline_candidates collection so screened and not-invited rows can be reviewed. Creating an invitation copies the relevant role/settings/ATS snapshot into a scheduled session; the original PDF is retained only for an invited session. This storage supports the recruiting workflow and is not an automatic training or continuous-learning feed. Deploy the owner-scoped Firestore rule before using persisted rows.

Ignored local .benchmark-runs files contain allowlisted synthetic inputs/outputs, settings/ hashes, timestamps, usage metadata and failures. Credentials and complete SDK request/response objects are not serialized. Missing keys produce no provider record. There is no automatic retention/deletion job: the owner manages scratch files and keeps selected fictional evidence for the case study.

Review form selections survive reload. Completed human records include source IDs/ hashes, transcript/model/rubric versions, judgments, alias, notes and timestamp. Browser storage and private host records have no automatic retention/deletion schedule. They are not anonymous analytics, model-training consent or tamper-proof audit evidence. Public examples should remain fictional and use a non-identifying alias. Export a reviewed record intentionally before reusing it in an evaluation dataset.

Future learning loop: not implemented

The separate candidate interview room records session-integrity metadata and recovery checkpoints after a candidate notice. Recovery checkpoints contain answers, partial questions, retired exchanges, remaining time and interruption/recovery counts in browser localStorage, including after tab closure. Terminal sessions attempt a hosted save. These are personal/session-linked records, not anonymous analytics or training data. There is no automatic deletion schedule or cross-device recovery. Use synthetic content for testing; the guide documents access, retention, client tampering and production authorization gaps.

Aggregate issue categories, create synthetic examples demonstrating recurring problems, review them, and evaluate prompt/model changes on a separate set. Reviewer corrections require calibration and are not automatically correct training labels. Past hiring outcomes are not unquestioned ground truth.

Fine-tuning, real candidate reuse, and provider fallback are separate future decisions. Before real-data deployment, specify purpose/lawful basis, candidate notice, access controls, retention/deletion, processing regions, provider policies, relevant cross-border transfers, and handling of data-subject requests for stored data and derived models. Interview processing does not automatically authorize training reuse.

Removing names alone does not establish anonymisation: free text, career details, voice and linkable identifiers can remain identifying. The ICO guidance informs this design; applicable law depends on deployment.

For Indonesia, include Law No. 27/2022 in the jurisdiction-specific assessment. No exhaustive legal mapping or compliance sign-off has been performed.

An open model served by a remote API is cloud processing. A provider's no-training policy differs from data staying on the user's machine. Do not silently switch restricted data to a new provider.

Appendix K - Future feedback and privacy-aware learning

Supporting report included from the current repository documentation.

Updated: 2026-09-11. **Roadmap proposal only.** The feedback form is implemented; local PII processing, curated training datasets and fine-tuning are not. This proposal does not authorize exporting or training on existing candidate records.

Case study / PRD · Data governance

Goal and order

Use candidate experience feedback to identify recurring product problems, turn approved examples into regression cases, and evaluate improvements before changing a production model. Start with prompt, UX and orchestration fixes. Consider fine-tuning only when reviewed examples demonstrate a repeatable problem those changes do not solve.

1. Enhance the existing feedback loop

The form already collects overall experience, interviewer clarity, transcription accuracy, question relevance, technical reliability and an optional comment. Proposed additions are optional issue categories and a selected question/turn reference so the team can locate the reported problem. Do not attach complete transcripts to general analytics events.

Track response rate (submitted feedback / eligible completed sessions), each metric's count and distribution, recurring issue frequency, and verified resolution rate. Show sample sizes and language/model/version when cohorts are large enough to avoid exposing individuals. Missing feedback is not a negative rating. Exclude seeded demo records from product metrics.

Candidates report their experience; their satisfaction score does not establish whether a competency score is correct. Independent reviewers must inspect quotations, rationale and rubric levels when creating evaluation labels. Past hiring decisions and AI-generated scores are not automatic ground truth.

2. Proposed data flow

mermaid example

```

graph TD
    A[Feedback and authorized session history] --> B[Purpose and reuse permission check]
    B --> X[Exclude from learning dataset]
    B --> C[Minimal selected excerpts in private environment]
    C --> D[Local rules and local entity model]
    D --> E[Mask PII and generalize identifying context]
    E --> F[Normalize formatting without changing meaning]
    F --> G[Second local scan and authorized human review]
    G --> Q[Quarantine or exclude]
    G --> H[Versioned de-identified dataset]
    H --> I[Regression and prompt evaluation first]
    I --> J[Optional fine-tuning experiment]
    J --> K[Held-out quality and privacy evaluation]
    K --> L[Staged release with rollback]
  
```

A transcript has already been processed by cloud services in the current hosted interview flow. A later local sanitizer cannot undo that prior processing. The boundary proposed here protects **future dataset preparation/export**; it is not

a claim that the existing live interview is entirely local.

3. Local masking and normalization

Run the sanitization worker on an operator-controlled workstation or an isolated private server. A remote Ollama endpoint is still remote processing, even if the model weights are open. If using a private cloud server, document that boundary rather than calling it on-device processing.

A small CPU entity-recognition model plus deterministic recognizers is the first implementation candidate; a large generative LLM is not required. A locally hosted LLM may assist contextual detection if resources permit, but cannot approve its own output for export. Evaluate English and Indonesian recognition separately, including code-switching and local identifier formats.

Proposed controls:

- Detect direct identifiers: names, emails, telephone numbers, addresses, account/identity numbers, URLs, GitHub handles and employer/client identifiers where identifying or confidential.
- Inspect indirect identifiers: rare project descriptions, exact dates, locations, company/role combinations and unusual incidents. Generalize or exclude passages that remain identifying.
- Replace entities with within-record placeholders such as [PERSON_1], [ORG_1] and [EMAIL_1]. Avoid stable cross-record identity tokens. Exclude session IDs and identity mappings from exports; protect any necessary deletion/provenance mapping separately.
- Normalize whitespace, encoding and verified transport duplicates. Preserve speaker/turn boundaries, negation, uncertainty, chronology and technical details needed for evidence. Do not rewrite hesitant answers as confident ones or silently change numerical claims.
- Keep source-to-sanitized turn mapping in the restricted audit area. Validate dataset quotations against the sanitized text so the example and target still agree.
- Scan optional feedback comments as well as transcripts. Avoid raw audio in the initial learning dataset; a voice recording has different identification risks from text.
- Block export on processing failure, uncertain entities or unresolved review. No cloud fallback for raw sanitization inputs. Keep outbound access, logs, temporary files, access permissions and retention under explicit control.

For example, I built the fraud dashboard at Acme with Jane; contact jane@example.com could become I built the fraud dashboard at [ORG_1] with [PERSON_1]; contact [EMAIL_1]. Even that sentence may remain identifying in context; replacement alone does not prove anonymity.

Presidio is one candidate library because it supports combinations of rules and entity recognition and local execution. Its own documentation states that automated detection cannot guarantee finding every sensitive entity. Tool selection, languages and thresholds need validation before adoption. Sources: Presidio overview, deployment and limitations FAQ (checked 2026-09-11).

4. Curated datasets and optional fine-tuning

Interview participation or ordinary feedback submission does not automatically authorize training reuse. Before inclusion, document the reuse purpose and applicable permission, provide a separate understandable choice where required, and define retention, access and withdrawal handling. Use synthetic examples first when reuse is unavailable or the source cannot be safely de-identified.

Create distinct datasets for distinct goals:

Goal	Reviewed input and target	Evaluation
Better questions and interviewer behavior	Role/context, prior exchange and reviewer-approved next question	Relevance, unnecessary repetition, unsupported assumptions, no spoken planning filler
Better evidence extraction	Sanitized transcript/rubric and reviewer-checked quotations/levels	Quote grounding, abstention, schema validity, level agreement and subgroup/language errors
Better transcription	Separately governed audio/reference pairs	Word/character error and meaning-changing technical-term errors

Text-only history does not by itself fine-tune a speech recognizer. Do not mix transcription training into the first text-evaluator experiment.

Version the dataset, sanitization policy, reviewer decisions, prompt, model and rubric. Split by source candidate/session before building train/validation/test sets, and keep duplicate or derived examples in the same split. Keep a held-out set untouched by prompt tuning and model selection. Remove leakage through names, identifiers, near-duplicate answers or reference labels in model inputs.

Compare fine-tuning against the existing model and a prompt-only baseline. Measure quality and failures alongside latency and cost; no default score improvement is assumed. Fine-tuning should target grounded behavior, not higher candidate percentages or imitation of historical hiring choices. Promotion requires explicit human approval and a rollback path.

5. Gates before a learning release

Gate	Required evidence
Reuse eligibility	Documented purpose/permission and exclusion of ineligible/seeded records
Sanitization	Labelled synthetic PII tests in EN/ID, per-entity precision/recall and contextual identification review; no unresolved detected identifiers in an exported record
Evidence preservation	Review confirms masking/normalization did not change responsibility, uncertainty, technical meaning or quote alignment
Dataset quality	Reviewed labels, version manifest, candidate/session-separated splits and duplicate checks
Model quality	Held-out comparison against current/prompt-only baselines, including unsupported quotes, abstention, language errors and latency/cost
Model privacy	Test memorization/extraction and restricted canary examples; residual risk remains explicit
Lifecycle	Access/retention/deletion procedure, derived-dataset traceability and rollback/version retirement plan

Deleting a source row does not reliably remove its influence from a trained model. Resolve the response to withdrawal before training: identify affected dataset/model versions and determine whether retraining or retiring a version is necessary. Do not promise automatic model unlearning.

Thresholds and sample sizes must be set before an experiment, based on the intended use; they are not yet measured or approved. Local processing reduces transfer exposure but still requires secure storage, logs, access control and a checked export boundary. Call the output **de-identified or pseudonymized** until a stronger anonymity assessment supports a different claim.

MVP boundary

The current submission can describe this as a future improvement supported by an existing feedback form. No background collection, export, local model deployment or fine-tuning is enabled by documenting it. The next practical step is a small synthetic-only masking/evaluation pilot, after the current quest submission.

Appendix L - Current runbook

Supporting report included from the current repository documentation.

Updated: 2026-09-11. Specification · Progress. This runbook supersedes the archived setup instructions. See the bulk candidate pipeline implementation note for the CV-to-invitation flow.

See model refresh, private access, validation and remaining setup for the latest implementation and the production release record for the verified deployment. Firebase ownership rules, indexes and the `pipeline_candidates` match are deployed. English is the baseline and Bahasa Indonesia is the separate multilingual extension.

Host setup

1. Run `npm install` and `npm run dev`. Use `http://localhost:3000` consistently; Firebase authorization for localhost does not authorize 127.0.0.1. Add your actual public domain to Firebase authorized domains before testing it.
2. In your private `.env.local`, configure `OPENAI_API_KEY` with valid Realtime access. The production server uses encrypted Vercel values for `OPENAI_API_KEY`, `FIREBASE_SERVICE_ACCOUNT_JSON`, `DEMO_COOKIE_SECRET` and `NEXT_PUBLIC_APP_URL`. Optionally set `GOOGLE_GENERATIVE_AI_API_KEY` and `DEEPSEEK_API_KEY` for other evaluators. Restart the server after local changes. Never paste keys into a report, repository or chat.
3. As the verified administrator, use **Create an interview invitation** on Dashboard. Copy the generated code and send it privately. The script node `scripts/reviewer-invite.mjs create "Sprint reviewer"` remains available for local setup; its code is written to the ignored `.demo-state/reviewer-invitation.txt`. Redeem the code at `/reviewer`. Run node `scripts/reviewer-invite.mjs revoke INVITATION_ID` to revoke one. The invitation file stores hashes; its separate delivery file contains the code.
4. Production runs on Vercel serverless with Firestore as the authoritative reviewer/demo ledger; no writable `/var/task` filesystem is required. Set a stable random `DEMO_COOKIE_SECRET` of at least 32 characters, provider billing alerts/limits and monitoring for cleanup failures.
5. Keep Firebase configuration for the ordinary recruiter pipeline and the hosted ledger. Reviewer scheduling and records use server-side Admin SDK access and do not require a fake Firebase user. Never change Firebase rules to public access to bypass a failed acceptance check.

Access and limits

Mode	Credentials	Limits and storage
BYOK	Personal keys entered in Settings; temporary request headers, never written by backend	Provider-account limits; keys remain in browser storage until removed. Shared-device users should clear them.
Demo	Host keys remain server-side	Five starts per signed browser cookie per UTC day, eight-minute funding window, eight connections, bounded answer/evaluation inputs. Clearing cookies can evade the per-browser limit; the shared host cap is the backstop.
Reviewer	Server-validated signed invitation	Default 30 starts/day, 12 billable/record operations per minute, 500 usage units, seven-day expiry; hosted candidate journey only, one active lease per invitation. Revocation checked on requests and call cleanup.

Voice reserves one usage unit per funded minute; an evaluation reserves three. Unlimited active interview time has a 30-minute hosted funding window, displayed before use. Units cap operations, not exact currency cost. Firestore transactions back the hosted ledger; the signed cookie is an admission handle, not an identity proof. Server-authoritative audit, retention/deletion tooling and stronger abuse controls remain future hardening.

Configuration and flow

Personal setup, scheduling and Reviewer Mode use InterviewSetupForm and `interview-config-v2`. Voice + text is mandatory: the candidate speaks, reviews the transcript in the answer composer, edits it if needed, and explicitly sends it. The Additional competency rubric is empty by default; the evaluator derives job-related competencies from the job description and validated CV context. Recruiters can add an optional competency from the dropdown when they need an extra explicit lens. Duration is 5, 10, 15 minutes or Unlimited. Every new question/follow-up consumes a turn; evidence coverage is calculated independently. Structured mode follows core questions; Adaptive mode requests relevant follow-ups. Adaptive topicality is prompt-directed and still needs human testing.

The recruiter bulk pipeline at `/pipeline` reuses the ATS checker for multiple PDF resumes. It ranks candidates against one role brief, supports optional Top 5/10/20 preselection and manual checkboxes, then creates the same scheduled interview links as the existing flow. Candidate email is required for each link so Firestore can enforce candidate-level admission. The `/candidates` navigation view keeps screened, not-invited and invited candidates together with expandable details; deploy the `pipeline_candidates` Firestore rule before relying on persisted pre-invite records. The old `/interviews` route remains a compatibility view for existing records. The hosted `/demo` route now requires sign-in; it still uses server-side AI credentials and browser-scoped demo usage, while the private `/reviewer` invitation remains the constrained candidate journey.

The reviewer journey is intentionally narrow: the administrator creates an invitation, the recipient redeems it at `/reviewer`, completes setup (including an optional CV), completes the live interview, sees the evidence-based evaluation once, and finishes. The recipient has no recruiter dashboard, Evaluation Sandbox, Human Review controls or access to other candidates. The completed evaluation is saved in private server-side reviewer storage and loaded into the verified administrator's Dashboard. No raw CV PDF is stored in that ledger; only bounded parsed context is used for the interview.

Recruiter scheduling snapshots configuration and role context. CV parsing returns status, bounded text, pages, characters, warnings and failure reason; confirm the preview or explicitly continue without CV. Scanned PDFs have no extractable text; no OCR is included. GitHub metadata is ranked before retrieving up to three cleaned, bounded READMEs. Retrieval failure is non-blocking and repository ownership does not prove personal contribution.

All voice rooms share the turn controller. OpenAI WebRTC uses `/api/realtime` and `src/lib/realtime/service.ts`; optional Gemini speech also uses `/api/realtime/gemini`. Old token creation is retired. An initial failure offers Retry Connection, Return to Setup and Exit Interview. Recovery retains completed answers and replaces the interrupted question. Technical interruption and skipped answers never directly reduce competency scores. Browser focus/visibility does not establish cheating and cannot inspect other devices or another app's content.

Step-by-step acceptance

1. **No keys:** sign in, arrive at Dashboard; open personal setup and verify missing-key guidance prevents a paid voice start. Open `/interview` without context and verify recovery/navigation actions. No forced `/set up` redirect.
2. **Reviewer access:** create an invitation as the verified administrator, redeem it in a separate browser, and verify the recipient sees only setup. Complete a synthetic interview, confirm the recipient sees the evaluation once, and confirm the result appears only in the administrator's Dashboard. An incognito visitor without a code must not

access the journey. Test revocation with a disposable invitation, not the invitation used by an active reviewer.

3. **Configuration:** create Structured 5-minute and Adaptive 10-minute sessions; confirm custom questions, language, rubric and panels. Create a scheduled synthetic session, change setup afterward, reopen the link and verify the saved configuration.
4. **Bulk CV pipeline:** follow the CV pipeline validation checklist with the ten fictional PDFs. Confirm parsing/name/email extraction, descending ATS bands, Top-N preselection, manual checkbox changes, scheduled-link creation, configuration/ATS snapshots and /candidates details. Verify uninvited rows remain visible and are not given a link.
5. **CV/GitHub in a single session:** try a text PDF, scanned PDF and non-PDF. Check preview and explicit no-CV continuation. Try a relevant GitHub username, then an unavailable one; the latter must still permit entry.
6. **English voice baseline:** with valid credentials, acknowledge the rules and start. Say only "uh, hmm"; no new question should follow. Give a meaningful answer, pause briefly and continue; verify approximately 2.5 seconds of quiet before a response. Wait after question playback: gentle text at about seven seconds, repeat/skip choices at seventeen. Mute/unplug the microphone; technical silence must remain separate.
7. **Adaptive and budget:** supply an answer with unclear ownership, then answer the follow-up clearly. Confirm one relevant follow-up, no repeated answered detail, and closing when the turn budget is exhausted. Record failures; prompt instructions alone do not prove compliance.
8. **Recovery:** use the session-control checklist. Confirm fullscreen pause overlay, warning, termination thresholds and interrupted-question replacement. Also fail initialization intentionally, return to setup and verify configuration remains.
9. **Evidence:** run empty, filler-only, partial, clear, contradictory and injected transcripts. Empty output must be Not Assessed with no percentage. Quotes must exist in candidate turns. Strong partial evidence keeps its assessed quality while uncovered criteria remain Not Assessed.
10. **Human review:** this remains an internal, contextual evaluation workflow and is not exposed through the candidate invitation. When an administrator opens a candidate or synthetic report, select all judgments, type ID/notes, reload, submit, reload again. Judgments must persist and completed fields become read-only; export includes source/version/model/rubric/hash and timestamp. Browser checks cover this path with synthetic data.
11. **Providers:** record primary/active model, fallback reason and errors. BYOK fallback requires the explicit Settings checkbox; it may send the transcript to another configured provider. Provider diagnostics reflect real requests and become stale; no inference health probes are sent. Model failures should produce no saved assessment.
12. **Indonesian extension:** repeat the same scenarios in Bahasa Indonesia with corrected reference transcripts. Keep its results separate from English. A single bilingual evaluator is useful for a pilot but not inter-rater agreement.
13. **Release:** use the production release record, record the five-minute video and attach the three submission artifacts. Do not claim independent recruiter accuracy or fairness results that were not measured.

Engineer handoff

Panel sync fix (2026-09-10)

The code editor and whiteboard previously accessed `interview_code` and `interview_whiteboard`, which the repository's Firestore rules deny. Their listeners also lacked error callbacks. Both panels now use fields `code_workspace` and `whiteboard_workspace` on an existing `interview_sessions` document, after Firebase authentication. No rules were broadened or deployed. Demo, Reviewer Mode and unauthenticated rooms remain local; their panel work is not saved or shared, as indicated in the panel. Permission failures retain the candidate's open panel and display the same limitation. Pending saves are cancelled on sign-out, session change or unmount.

Validation: 13 targeted automated tests passed for local-session isolation, authentication, both panel writes, denied reads/writes, missing records, read-only viewing and save cancellation. These use mocked Firestore; deployed permissions still need a signed-in browser check. Start a scheduled interview with Code Editor, type a short snippet, wait two seconds and check `code_workspace` on that session. Repeat with Whiteboard and `whiteboard_workspace`. Confirm Demo/Reviewer panels say Local only and produce no Firestore permission error. Do not mark live syncing verified until that check passes.

Configuration: `src/lib/interview/config.ts`; turn policy: `turn-policy.ts`; connection orchestration: `src/lib/store/useInterviewStore.ts`; evidence validation: `src/lib/ai/evidence.ts`; provider adapter: `evaluation.ts`; access/usage: `src/lib/access/reviewer.ts` and `src/lib/demo/ledger.ts`. Human reviews persist in the scoped Firestore ledger for recruiter/invitation records; local drafts remain browser-local. No training job or analytics export consumes these records automatically.

Run `npm test -- --runInBand, npx tsc --noEmit, npm run build, and npx cypress run --spec "cypress/e2e/revision.cy.ts, cypress/e2e/reviewer-demo.cy.ts, cypress/e2e/integrity.cy.ts"`. The revision browser test requires a valid local invitation and writes clearly synthetic local records. It does not call a paid model.

Before real hiring: design server-authoritative admission/configuration and auditing, scoped candidate tokens, retention/deletion tooling, accessible accommodations, validated scoring and independent human oversight. Existing browser checkpoints and reviewer-uploaded records are client-reported evidence, not tamper-proof audit records.

Appendix M - Five-minute demo script

Supporting report included from the current repository documentation.

Status: script ready; video not yet recorded. Target duration: **5:00**, including real interface actions and a short interview excerpt. Use English narration for the quest; Bahasa Indonesia may be a separately labelled optional clip.

Case study and PRD · MVP audit

Before recording

- Use the deployed app and the fictional CV batch/JD. Have one prepared completed record available for the result segment.
- Sign in beforehand. For a candidate walkthrough, use a separate browser profile and an invitation tied to an email you can actually access. The fixture's example.com addresses are not usable login accounts: edit one demo candidate's email to your test account before creating its link.
- Capture a short real English interview exchange in advance if necessary. Label it "Recorded prototype session"; a seeded dashboard score must say "Illustrative seeded record."
- Keep real names/emails, secrets and usable invitation codes out of the public recording. Show copying a link without publishing the full token.
- Record the parts separately, then combine them in the order below. Keep the candidate-session capture in its own window; switching tabs while an interview is active may trigger the disclosed integrity controls.
- Rehearse with a timer. Narration is deliberately short to leave time for clicks, AI playback and the answer draft. Do not read every table or model name aloud.

Part 1 - Problem and user (0:00-0:30)

Show: title and role brief in Pipeline.

Say:

"InterviewMate helps a recruiter move from a batch of CVs to interview evidence they can inspect. The problem is the handoff: checking resumes, preparing separate invitations, and then reconstructing what each candidate actually demonstrated. I started with desk research and an existing interview app, then connected these steps into one working prototype."

Part 2 - CV intake and selection (0:30-1:20)

Show: paste JD, choose fictional PDFs, then show completed ranking. A cut over processing should be labelled, not presented as a measured speed result.

Say:

"I enter the role once and upload several fictional CVs. The app extracts names, email addresses and resume text, then scores each CV against the same job description. I can inspect and correct the extracted information. This ATS score uses keyword and experience heuristics, so it is a screening aid. I can select the top five, then change any checkbox. The recruiter decides who gets an invitation."

Action: inspect one strong and one weak match; change one checkbox.

Part 3 - Invitation setup (1:20-1:50)

Show: Continue to Step 4, selected candidates, English, duration/turn budget, empty additional rubric, copy link.

Say:

"The selected candidates move to a separate scheduling page. I set the interview language, duration and question budget. Additional competencies are optional. The invitation keeps a snapshot of the role, settings and ATS result. The app creates a link for each selected candidate; sharing the link is still the recruiter's action."

Part 4 - Candidate interview (1:50-2:55)

Show: authenticated candidate setup, one complete AI question, spoken answer appearing in the composer, one edit and Send. Reserve about 30 seconds here for actual audio and interaction.

Say before the excerpt:

"The candidate enters their own session. Voice and text work together. Their speech becomes an editable draft, and they choose when to send it. That gives them control when transcription gets a technical term wrong."

Example candidate answer, if appropriate to the actual question:

"I built the React filters for an order search page. I added loading and empty states, and wrote tests for filters that returned no results."

Say after Send:

"Sending commits the answer and clears the draft. The next question can follow the role and the evidence still missing. Recovery preserves completed answers when a technical interruption occurs."

Do not stage an answer unrelated to the question. Show recovery only if there is time; it is not necessary to demonstrate every feature.

Part 5 - Evidence, history and feedback (2:55-3:40)

Show: clearly labelled completed record; ATS, evidence score, coverage and one exact quotation. Briefly show feedback.

Say:

"The result separates the resume's ATS score from interview evidence. Each competency has a level and a quotation from the candidate's answer. The score out of one hundred summarizes supported evidence across the rubric; missing evidence affects coverage. It is not a hiring recommendation. The recruiter can open the full record, while the candidate can leave feedback on clarity, transcription, relevance and technical reliability."

Part 6 - AI logic and testing (3:40-4:25)

Show: AI diagram and evaluation table in the case study.

Say:

"The stages are explicit: parse the CV, calculate ATS matching, transcribe speech, run the interview, then generate and validate structured evidence. I used authored synthetic transcripts, ten fictional CVs and my own live self-tests. Reported problems led to transcript, turn-taking, layout and persistence fixes. The recorded release passed one hundred ninety-one automated tests. This is prototype validation with one builder; recruiter time savings and

independent model accuracy are not measured."

Part 7 - Handoff and close (4:25-5:00)

Show: PRD source map, requirements and production URL.

Say:

"The handoff includes the workflow, acceptance criteria, AI input and output, scoring formulas, storage boundaries and error behavior. The app is deployed with server-side credentials. Next, I would measure manual versus assisted review and ask an independent recruiter to check the evidence levels. The MVP is ready to demonstrate the complete recruiting workflow, with those limitations documented."

Final checks and submission

Export one approximately five-minute video. Check that audio, text and score labels are readable; distinguish live/recorded sessions from seeded examples. Open the video link as someone without your account. Submit the app URL, case study/PRD and video together.

Video URL: not recorded.